

CIS 5800

Machine Perception

Instructor: Lingjie Liu

Lec 6: Feb 10, 2026

Recap:

Application Example: Using homographies to measure lengths!

- Homographies (i.e. World Plane $\rightarrow$ Image homographies) allow us to measure lengths in the world plane of interest.



- We will discuss this in detail in today's class.

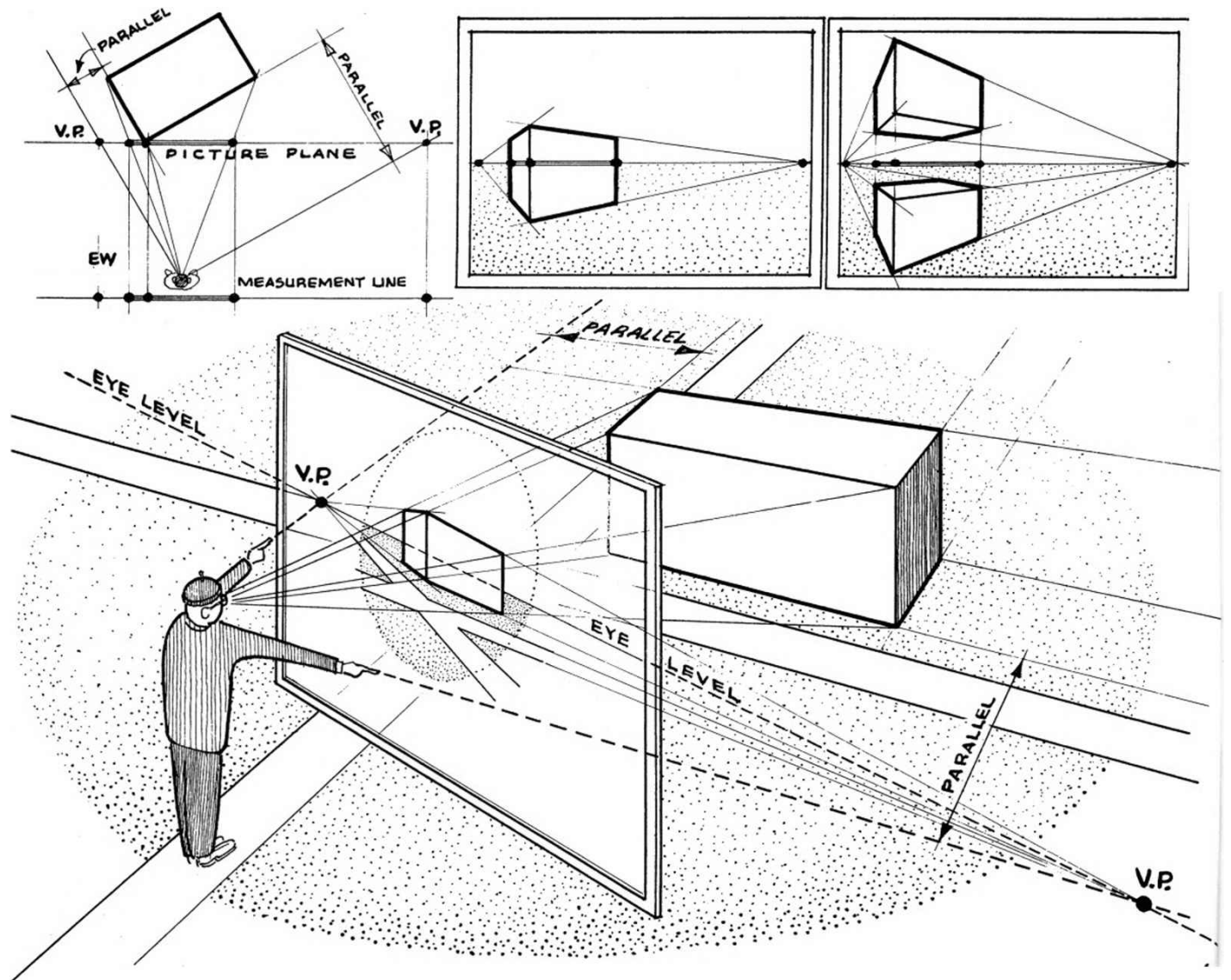
Recap:

How Artists Find Vanishing Points

Find VP of a world line by:

- Standing at “camera center”.
- Holding arm out parallel to the world line.
- Noting its intersection with the “canvas” or image plane. i.e. the arm represents the light ray.

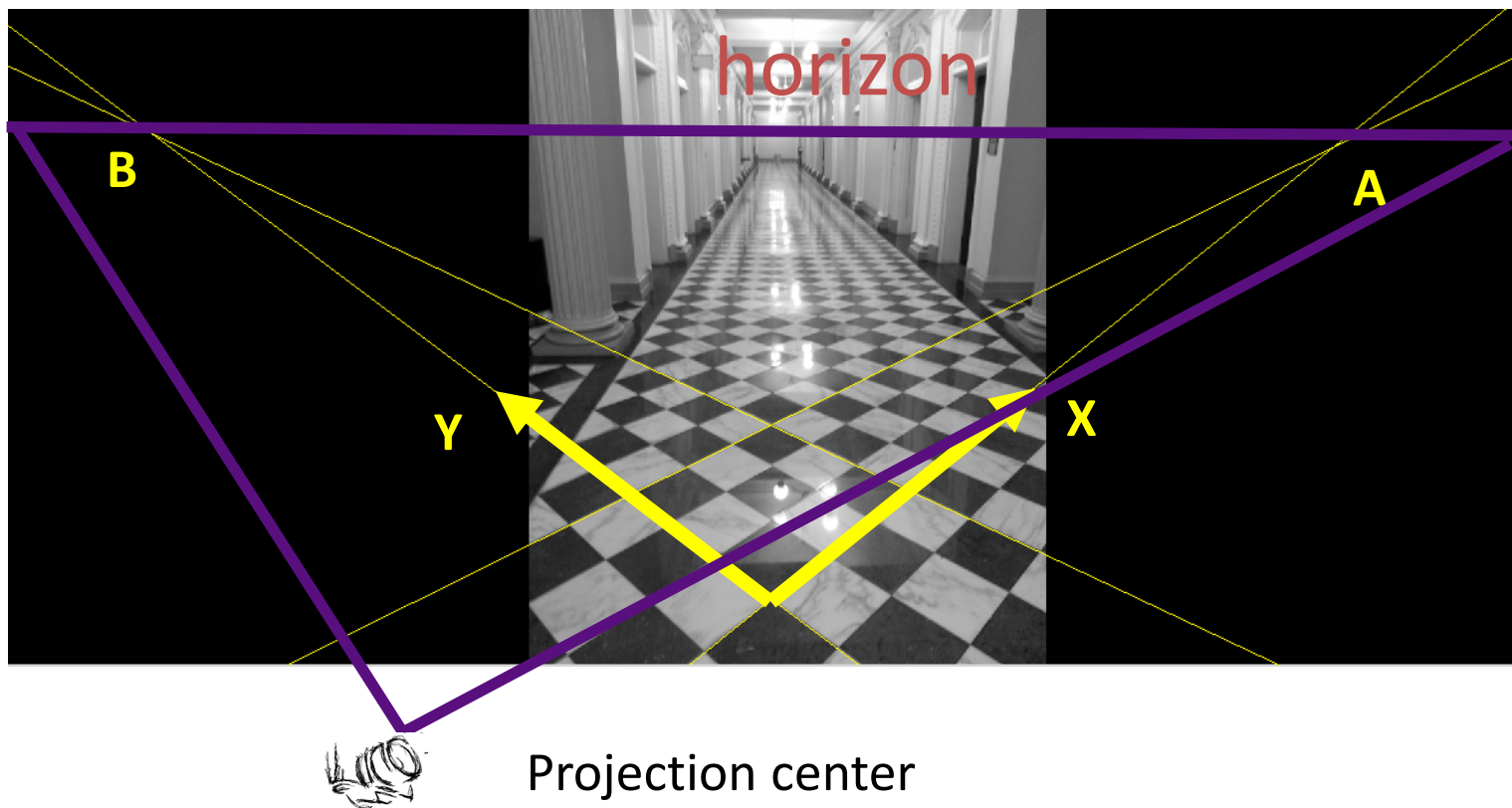
“Vanishing rays of a world line” (camera rays through the VP) are just rays parallel to that line, passing through the camera center.



Recap:

Vanishing rays/planes through the camera center are parallel to the world lines/planes

So, the horizon plane is parallel to the ground plane
and hence $h_1 \times h_2$ is the normal to the ground plane!



Horizon plane =
Vanishing plane =
Viewing plane

World plane //
vanishing plane

World plane = Ground
plane in this case

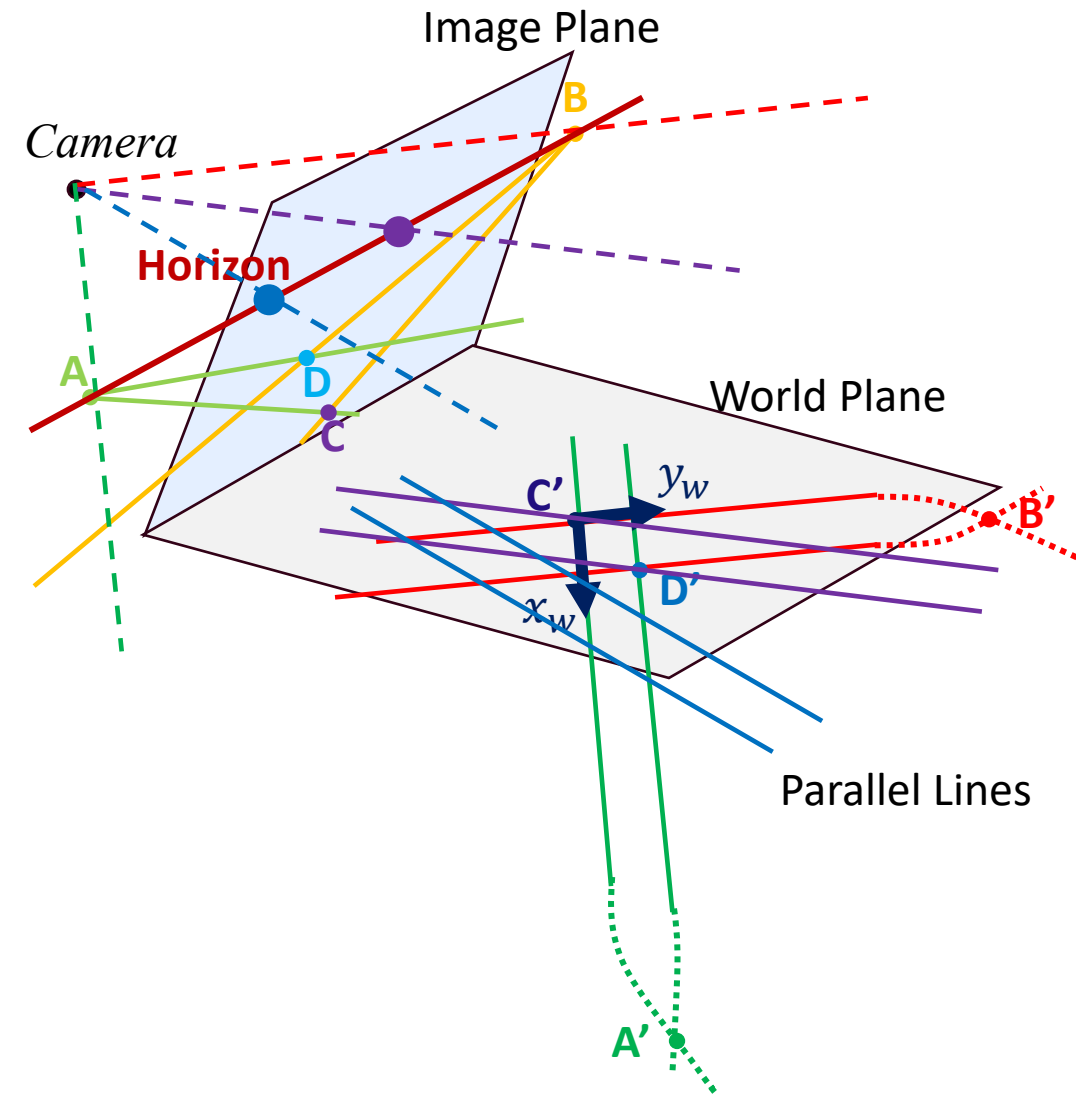
Recap:

Projecting the line at infinity to compute the horizon

Points at infinity in the world plane look like $(X, Y, W = 0)^T$

The “line” connecting them is $W = 0$, the “line at infinity”. The image of this line is the horizon, which contains all vanishing points. Expressed in world plane $\mathbb{P}^2$, this line’s coefficients are $(0,0,1)^T$.

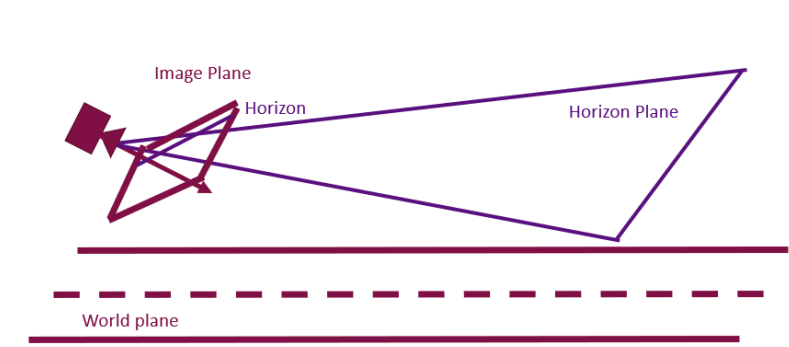
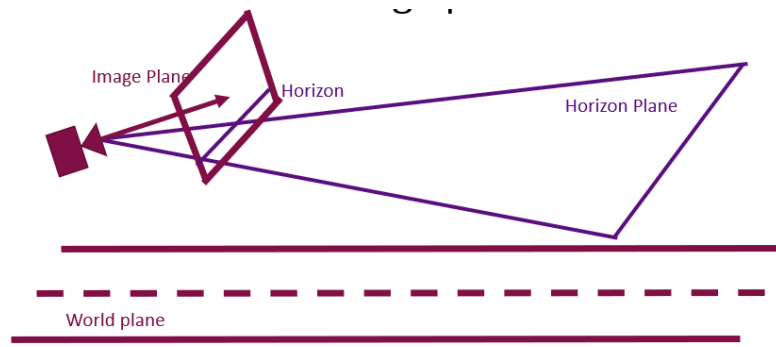
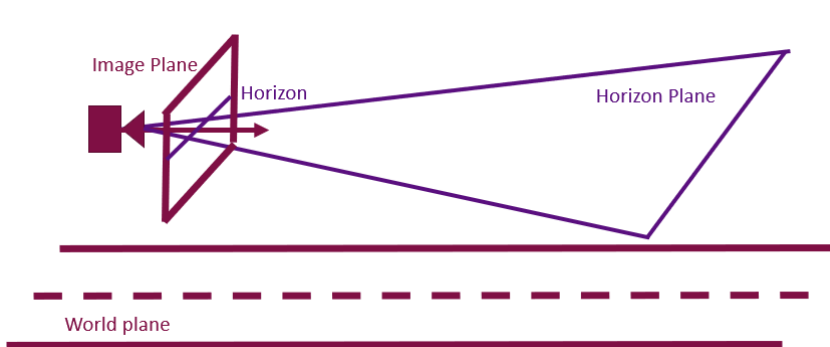
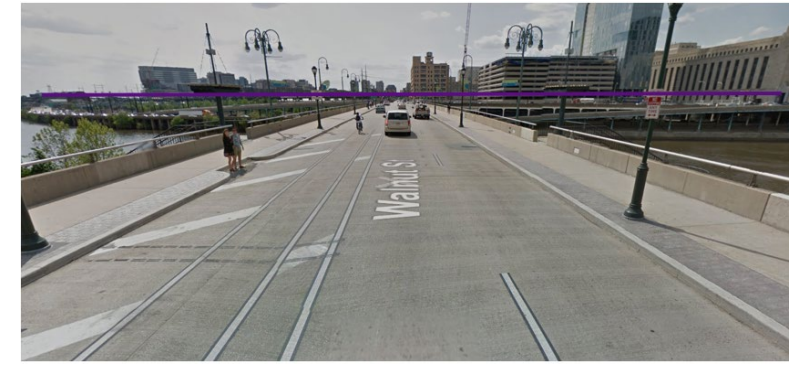
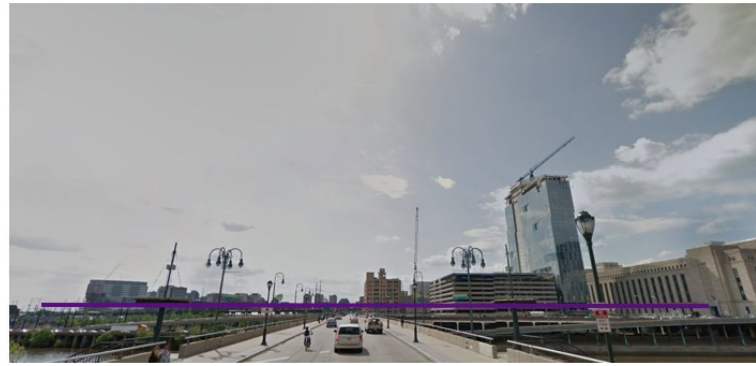
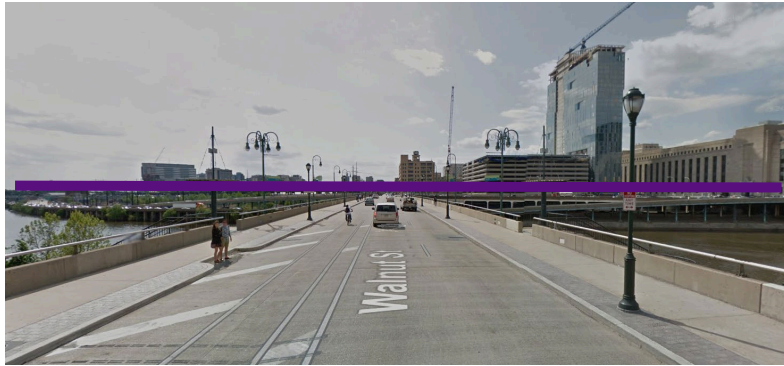
So if we could find the projection of this line, we could find the horizon



Recap:

Horizon gives complete info about how camera is oriented w.r.t. world plane*!

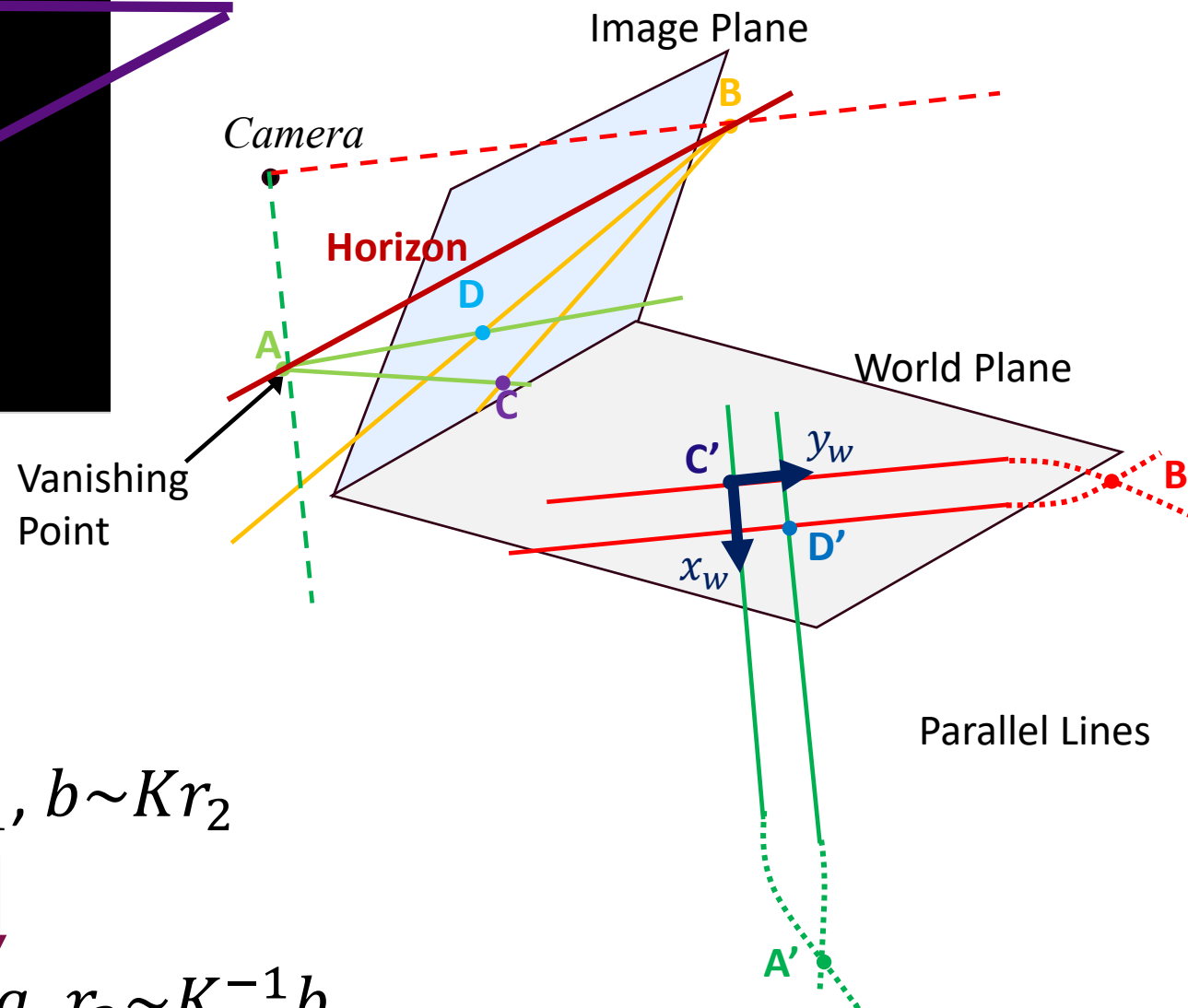
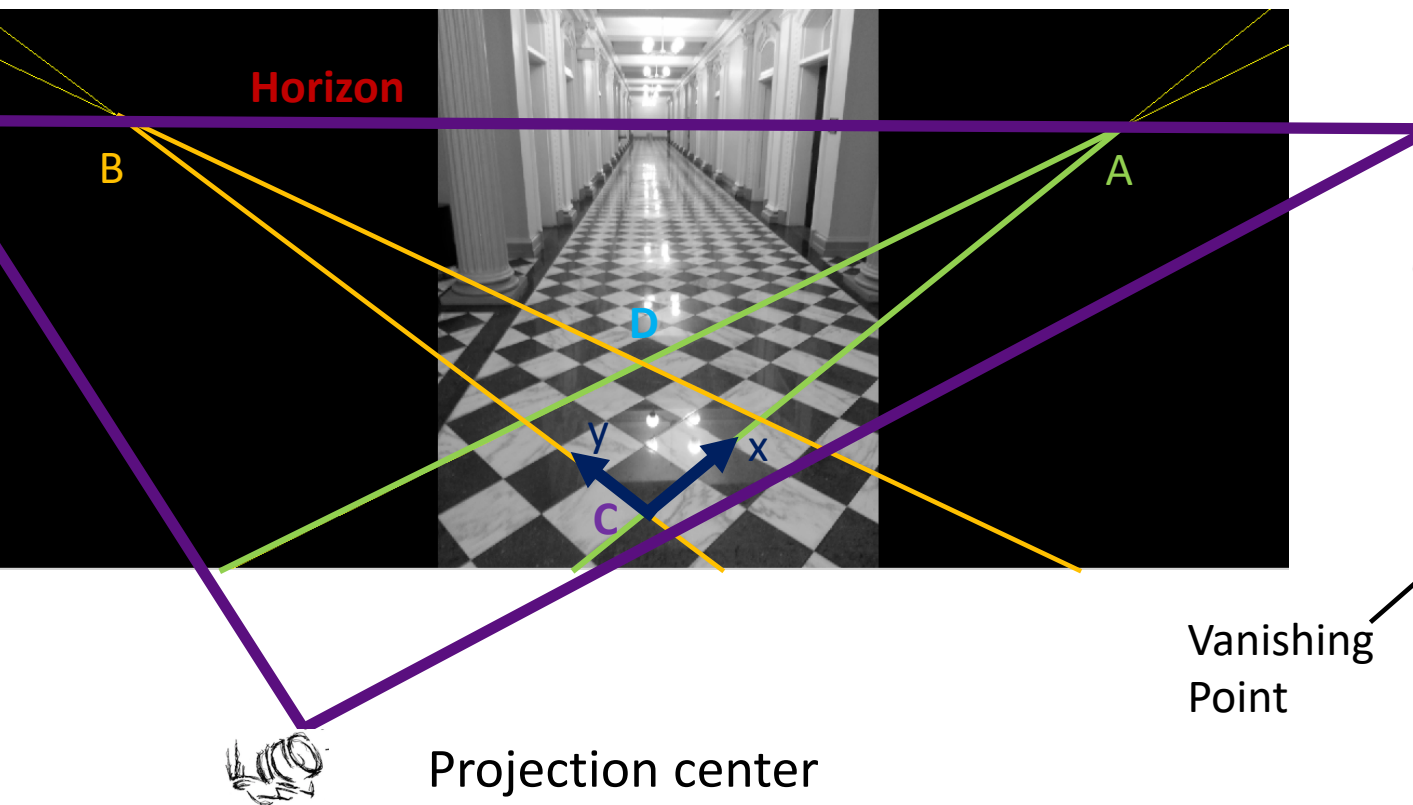
Thumb rule: “If horizon is horizontal & central, camera is correctly vertical & principal axis is parallel to world plane**!”



*caveat: assuming known K

**caveat: assuming that principal axis passes through image center, and camera axes are horizontal. (usually approximately true)

Recap:



$$H \sim K(r_1, r_2, T)$$

$$H \sim (a, b, c)$$

$$a \sim Kr_1, b \sim Kr_2$$

$$r_1 \sim K^{-1}a, r_2 \sim K^{-1}b$$

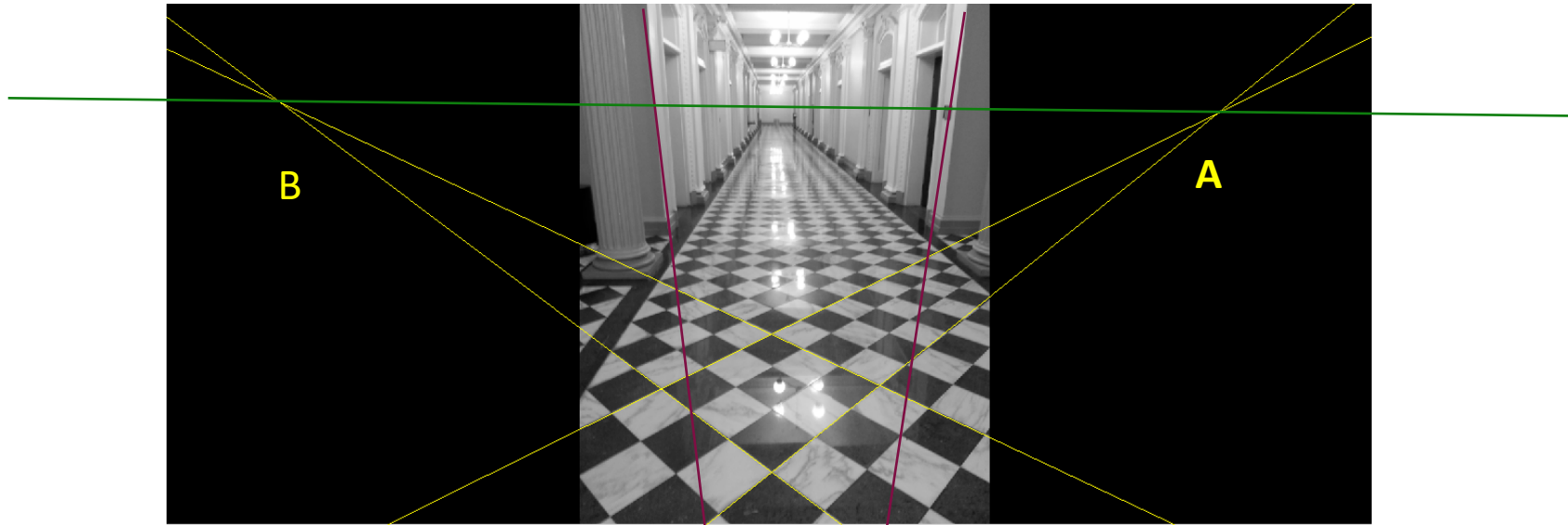
Recap:

- We can get the rotation of camera from 2 orthogonal vanishing points on a plane, **assuming known intrinsics matrix K** .
- Turns out, you can also get *intrinsics* from vanishing points.

$$\begin{array}{l} H \sim K(r_1, r_2, T) \\ H \sim (a, b, c) \end{array} \begin{array}{l} \nearrow \\ \nearrow \end{array} a \sim Kr_1, b \sim Kr_2 \longrightarrow r_1 \sim K^{-1}a, r_2 \sim K^{-1}b$$

How to compute intrinsics K from vanishing points

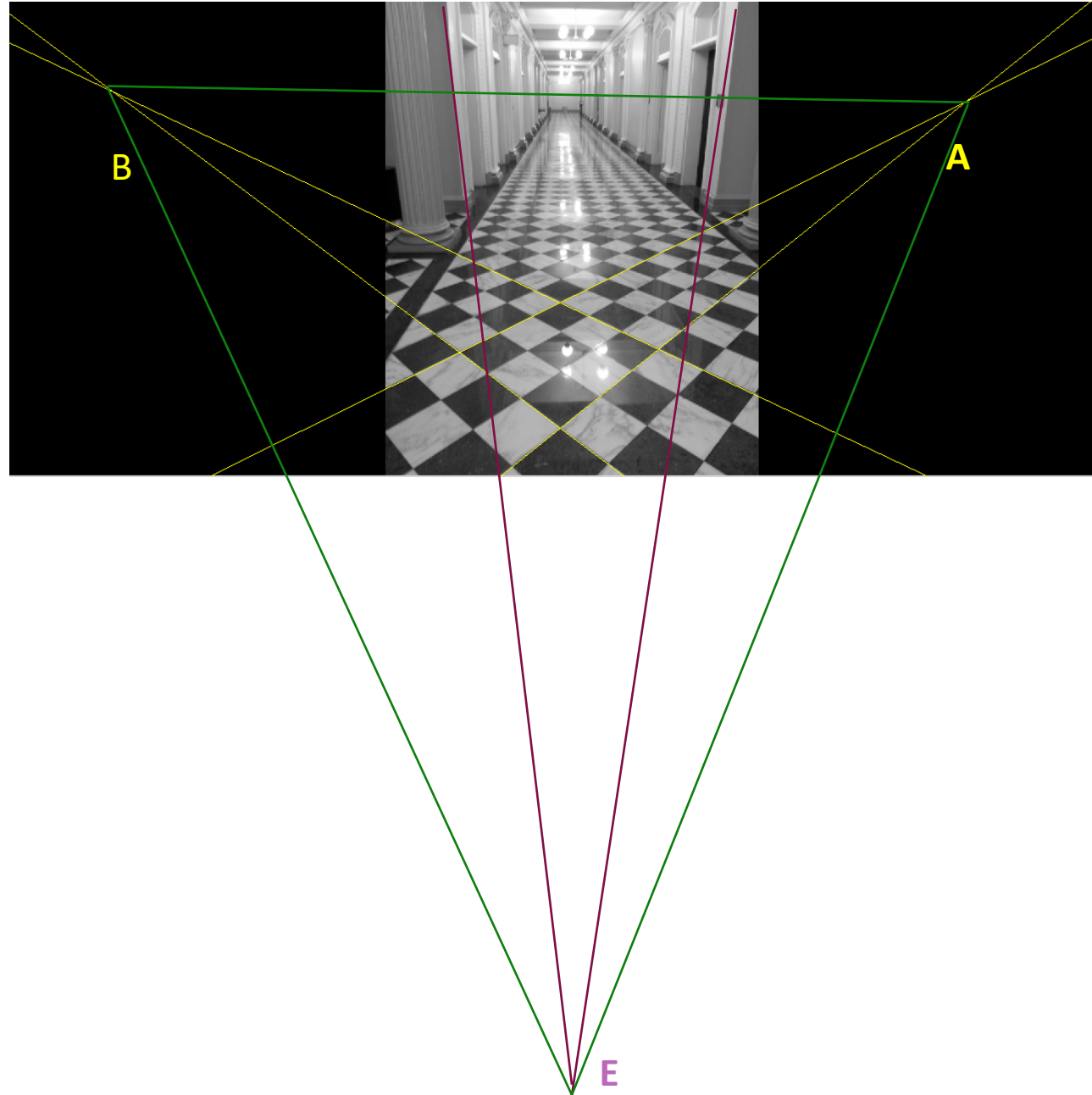
Line connecting AB is the horizon!



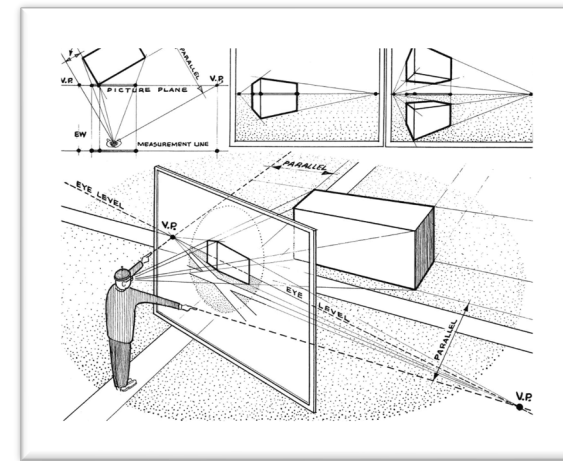
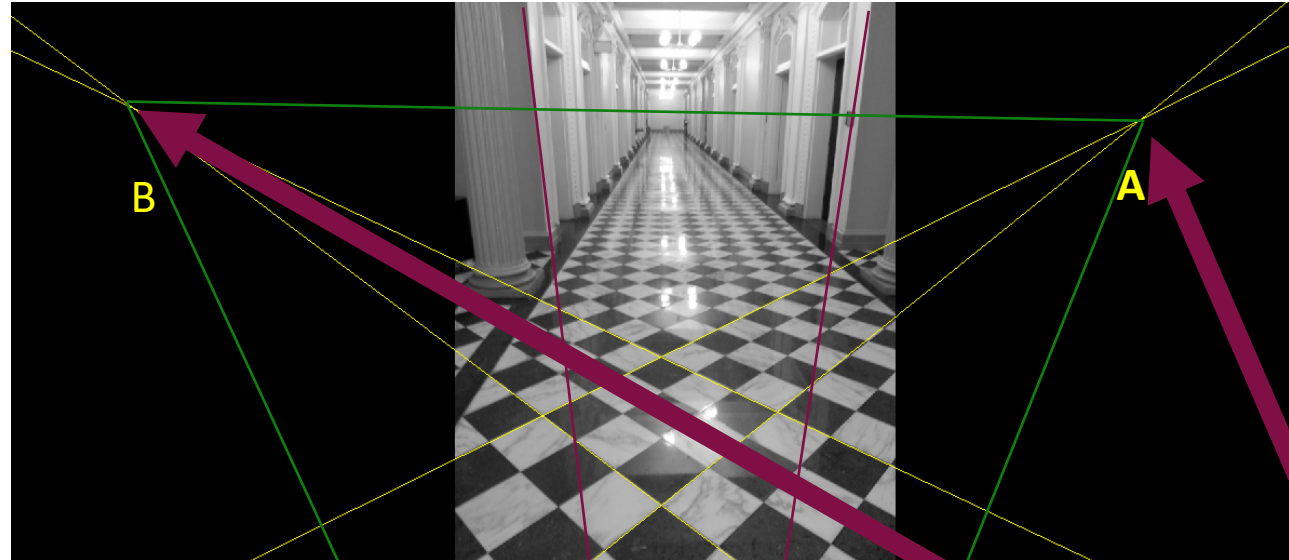
Remember that the horizon gives us the orientation of the ground plane with respect to the camera!

E

E is the vertical vanishing point!



Let's look at ABE as a tetrahedron OABE including the projection center

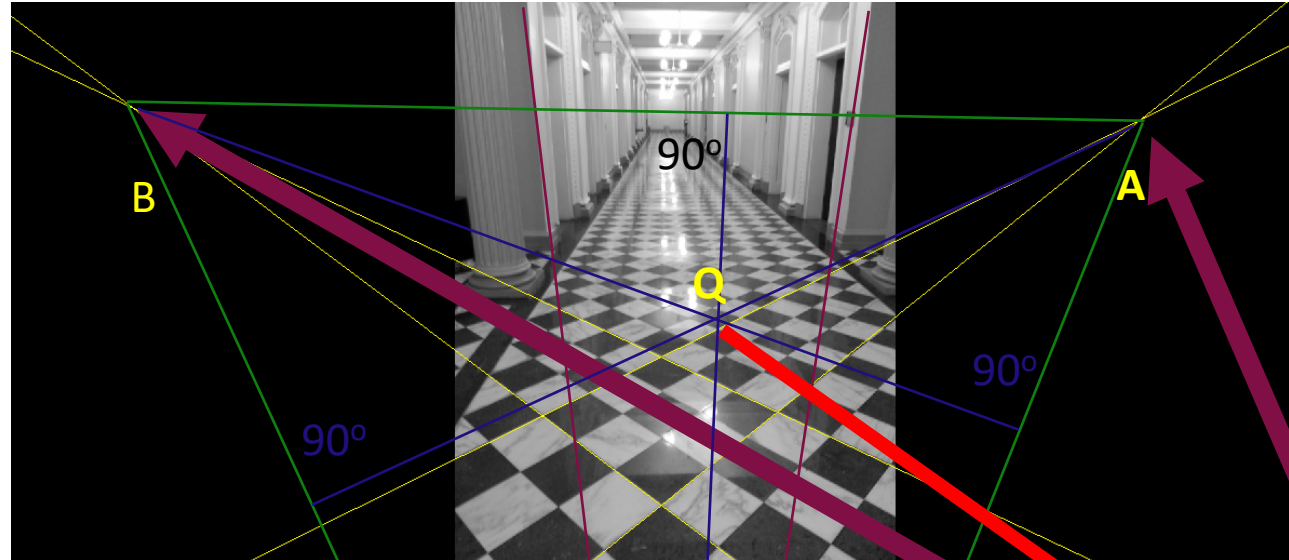


90°
90°
O is Projection
center

Tetrahedron vertex angles are 90 degrees

E

Let Q be the orthocenter of the triangle ABE

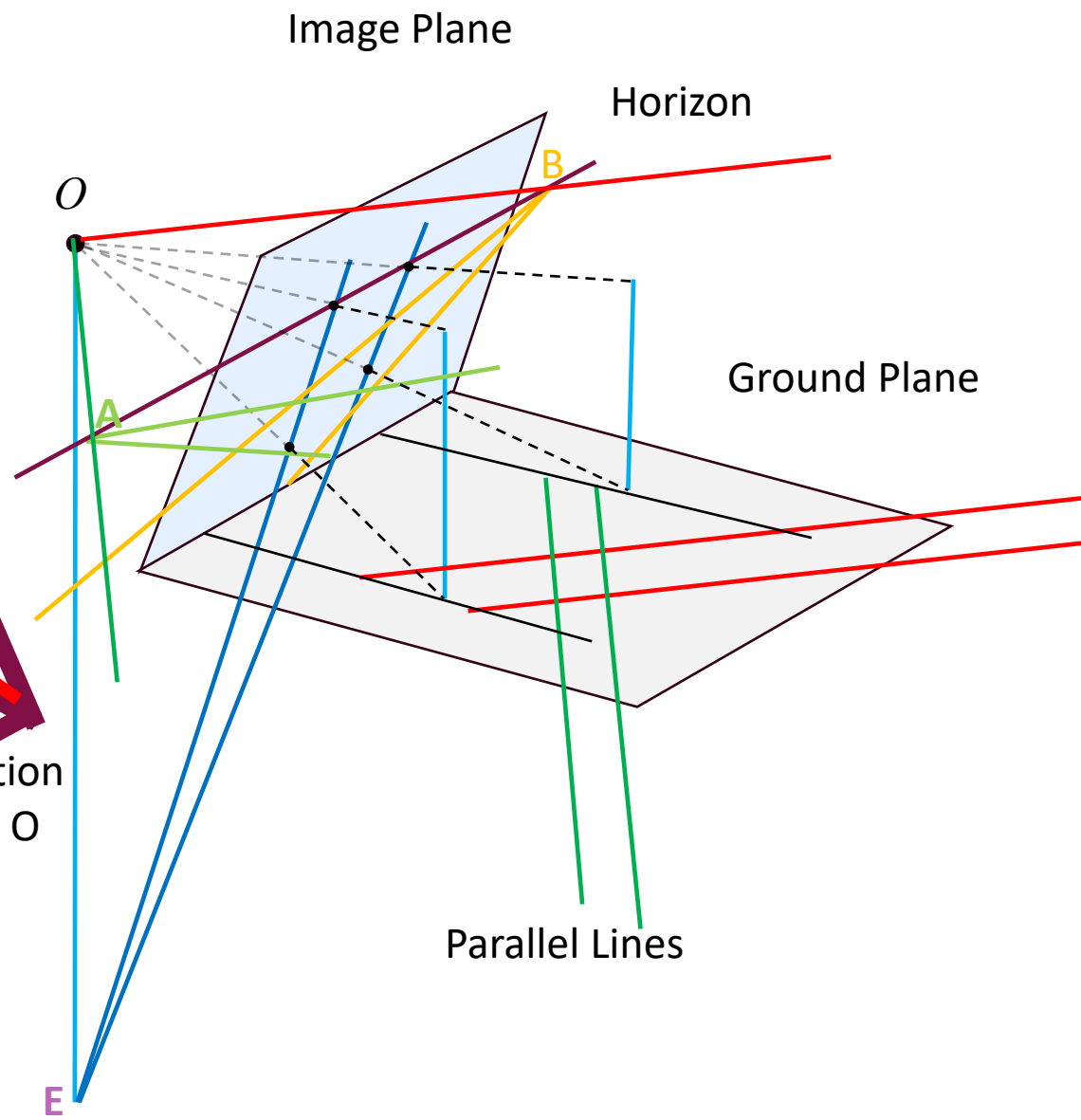
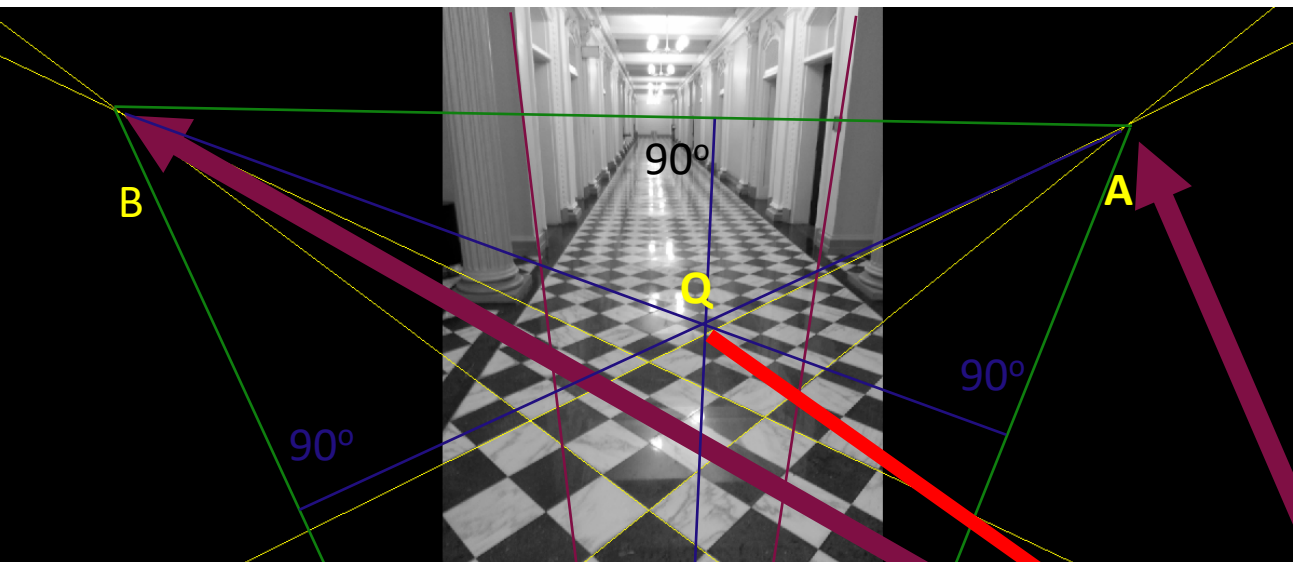


Theorem from Euclidean Geometry:

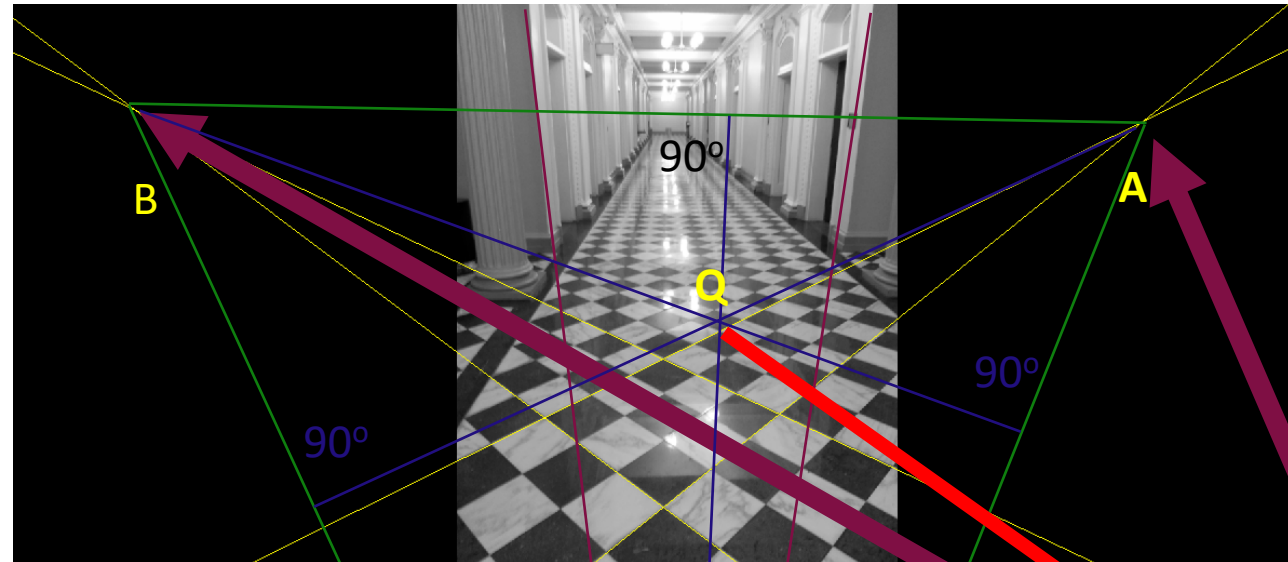
If Q is the orthocenter of ABE and all three angles AOB , BOE , and EOA are right angles, the OQ is perpendicular to ABE plane!

E

Projection
Center O



Let Q be the orthocenter of the triangle ABC



Projection
Center O

Theorem from Euclidean Geometry:

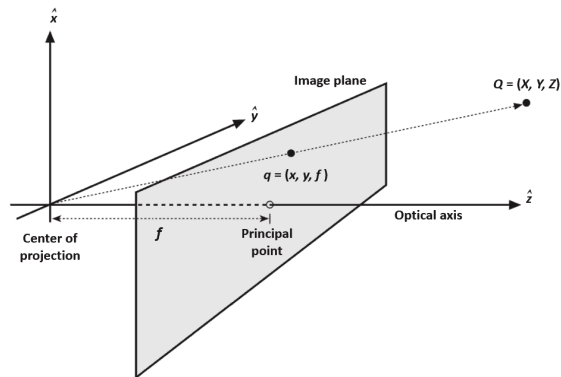
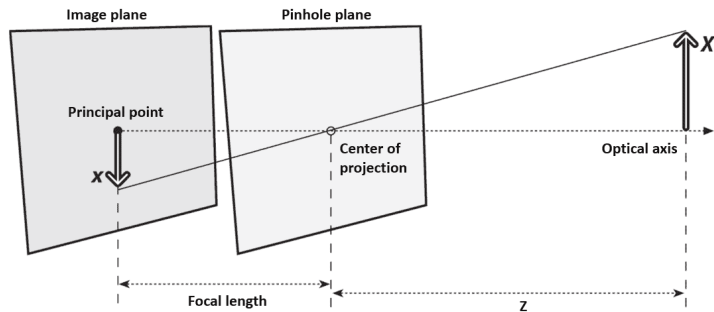
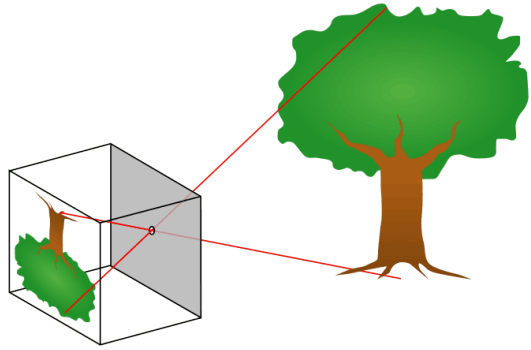
If Q is the orthocenter of ABE and all three angles AOB, BOE, and EOA are right angles, the OQ is perpendicular to ABE plane!

OQ is the principal axis and ABE is the image plane, hence, Q is the principal point

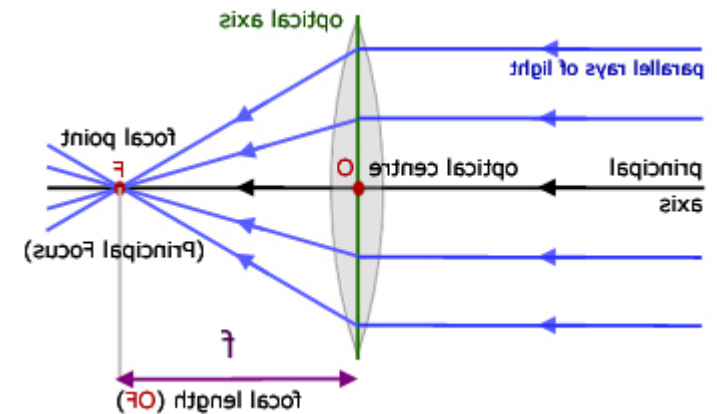
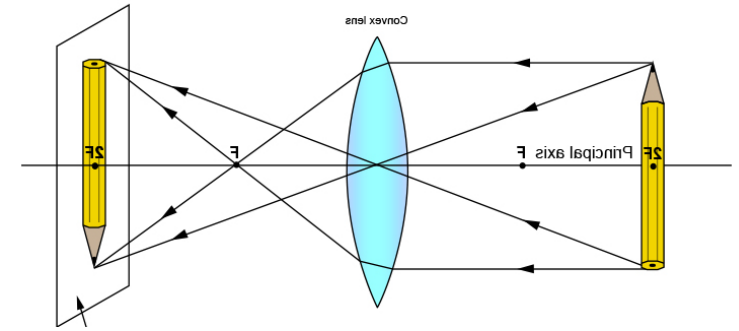
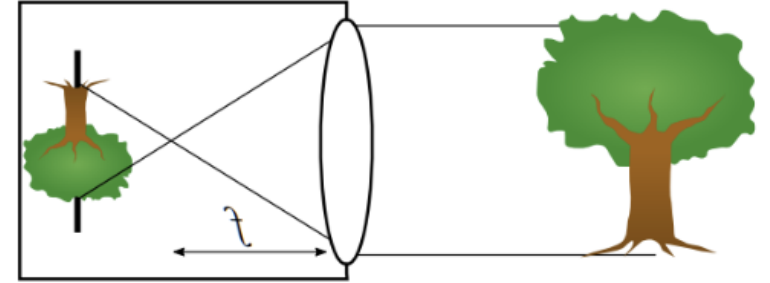
E

Side Note: Principal Axis/Optical Axis, Principal Point

Pinhole Camera:



Lens Camera:



Bonus proof: Perpendicular from camera center is the orthocenter of VPs

THEOREM: The image center (u_0, v_0) is the orthocenter of the triangle formed by the projections of three orthogonal vanishing points.

PROOF: See figure 2. Let $C = (u_0, v_0, 1)$ denote the homogeneous coordinates of the image center: it is defined as the intersection of image plane $V_1V_2V_3$ with the optical axis, which is the line through O and perpendicular to $V_1V_2V_3$.

$$OC \perp V_1V_2V_3 \Rightarrow OCV_1 \perp V_1V_2V_3$$

$$\Rightarrow OCV_1 \perp \text{any line contained in } V_1V_2V_3$$

By B: every plane containing OC is $\perp$ to $V_1V_2V_3$

By C: $V_2V_3 \perp CV_1$, so $OCV_1 \perp V_2V_3$

In particular $OCV_1 \perp V_2V_3$. Moreover, $OV_1 \perp OV_2$, therefore $OCV_1 \perp OV_2V_3$.

When two planes are perpendicular, their intersections with a third plane are also perpendicular. Therefore, the intersection of OCV_1 with $V_1V_2V_3$ is perpendicular to intersection of OV_2V_3 with $V_1V_2V_3$.

In other words $V_1C \perp V_2V_3$. A similar reasoning leads to $V_2C \perp V_3V_1$ and $V_3C \perp V_1V_2$, therefore C is the orthocenter of $V_1V_2V_3$.

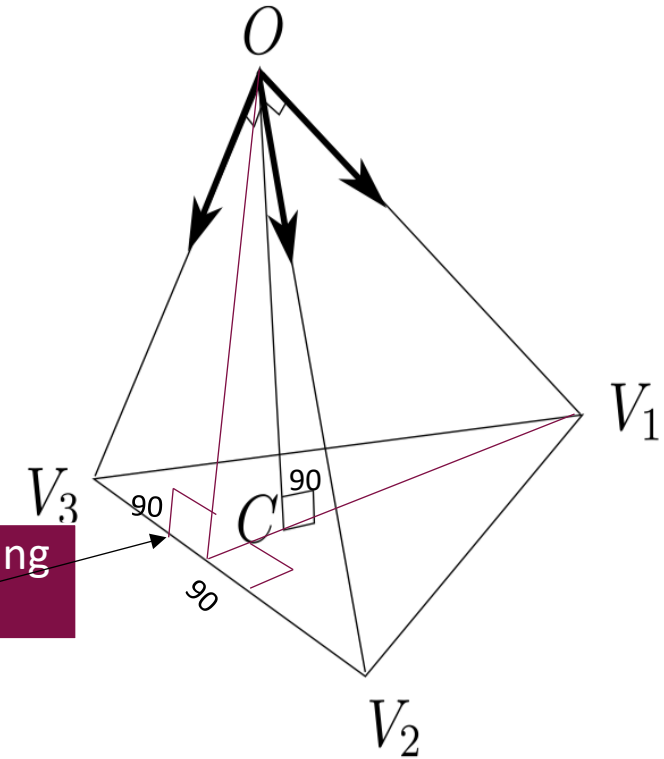


Figure 2: Three orthogonal vanishing points and image center.

We used the fact that the two non-parallel lines OV_2 and V_2V_3 contained in the plane OV_2V_3 are perpendicular to OCV_1 , therefore $OV_2V_3 \perp OCV_1$.

We found the principal point! What about the focal length ($f=OQ$)?
Can it be computed from A,B, and E ?

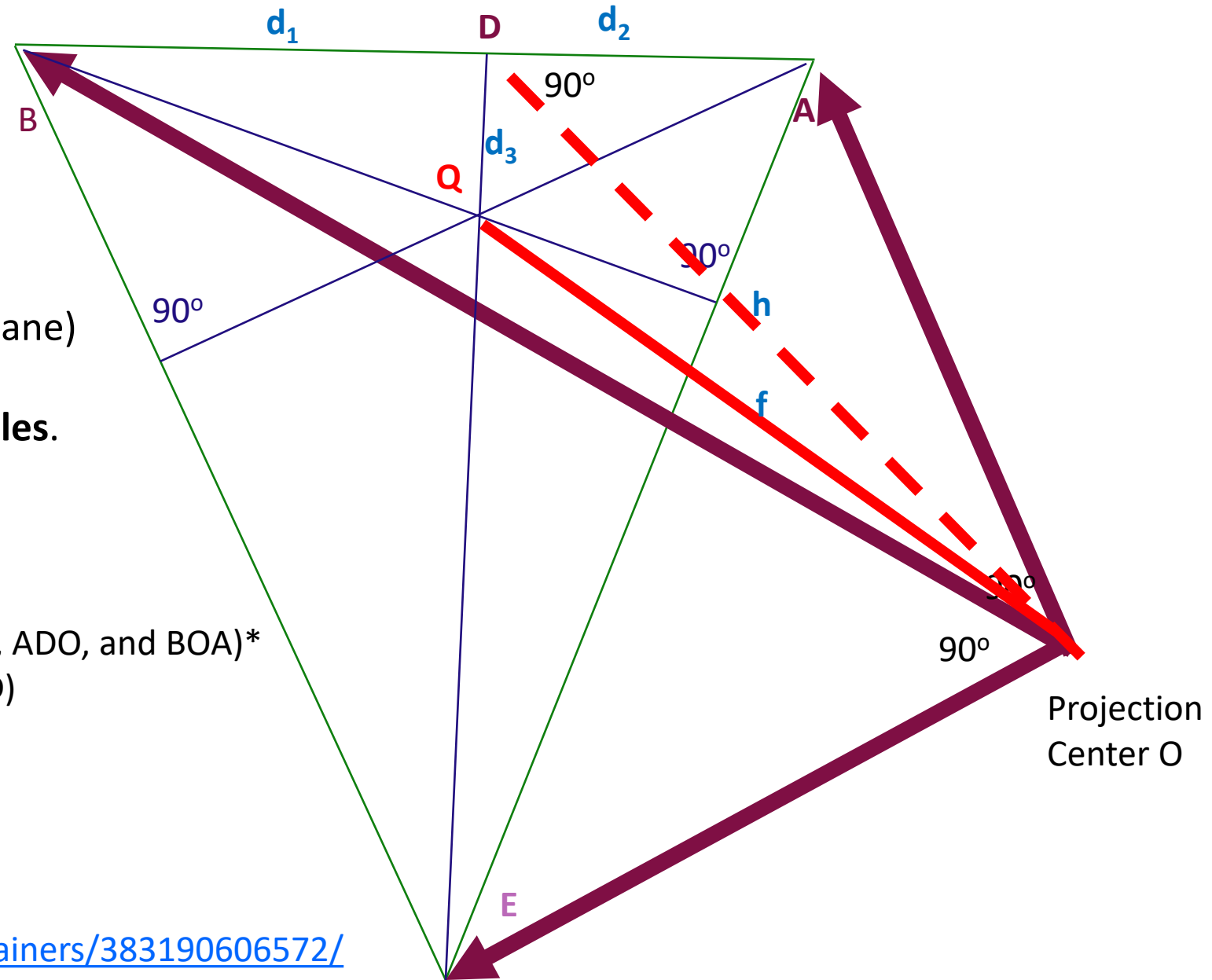
OQ $\perp$ image plane.
So plane **OQD** $\perp$ **image plane**

AD $\perp$ ED (int. of OQD & img plane)
So, AD $\perp$ OD
So, **BDO**, **ADO** are right triangles.

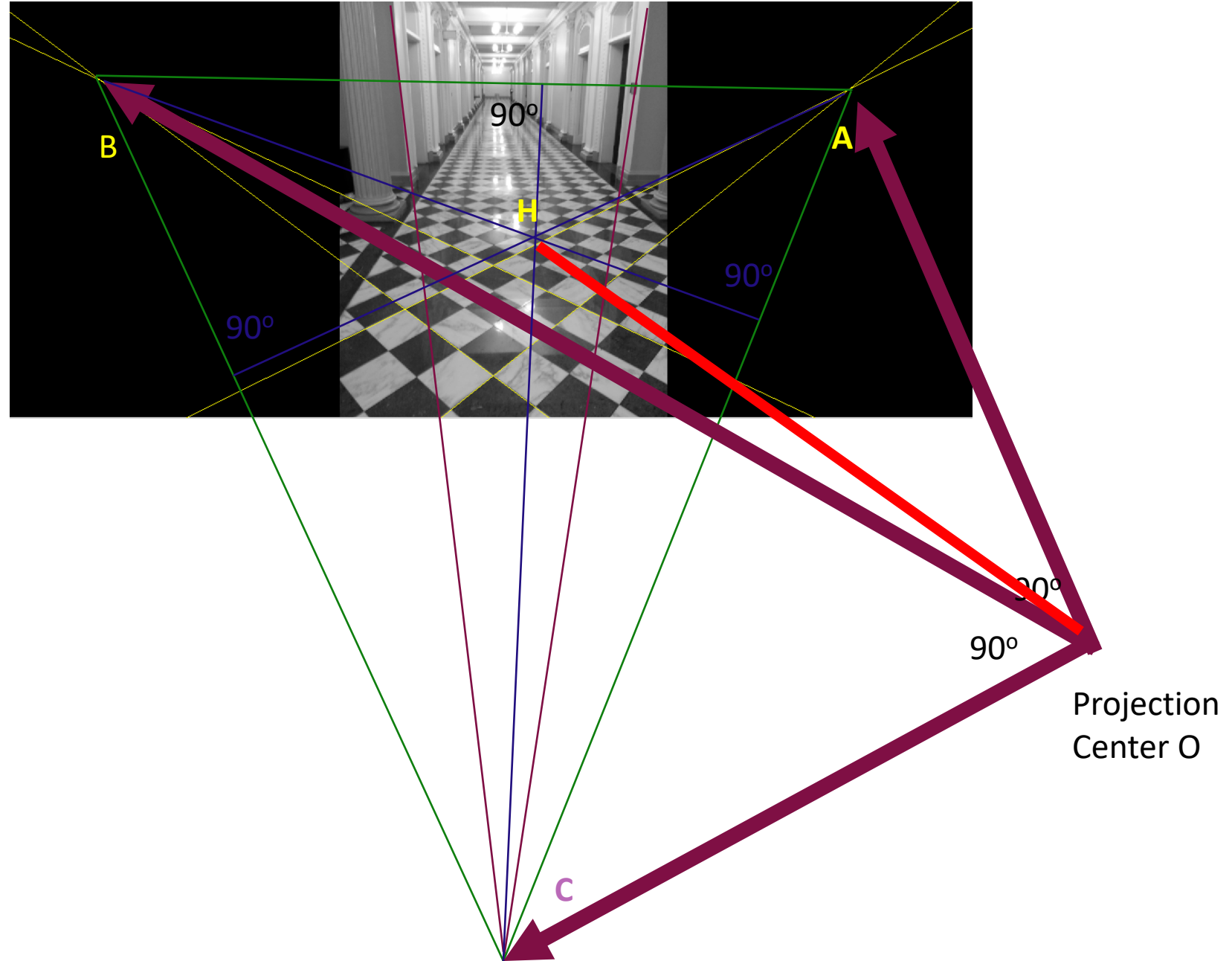
$h^2 = d_1 d_2$ (apply Pythagoras to BDO, ADO, and BOA)*
 $f^2 + d_3^2 = h^2$ (Pythagoras again to OHD)

Hence $f^2 = d_1 d_2 - d_3^2$

* <https://www.nagwa.com/en/explainers/383190606572/>



Three orthogonal vanishing points (if none is at infty) allow computation of full intrinsics K (focal length and image center)!



Summary

- If we have 3 orthogonal VPs, we can get full intrinsics K (focal length and image center), and also extrinsics R .
 - What's missing? Just the translation t .
 - And that information is not contained in VPs, because camera translations don't affect the VPs!
 - Which is why, when we found homographies (that do contain full information about translation), we used more than just VPs.

Cross Ratios & Length Measurements from Single Images (“Single View Metrology”)

Are lengths preserved under homography?



Obviously not.

What about length ratios?

Projection of a circle

From other perspective views, would the center of this circular clock face remain at center?



1:1



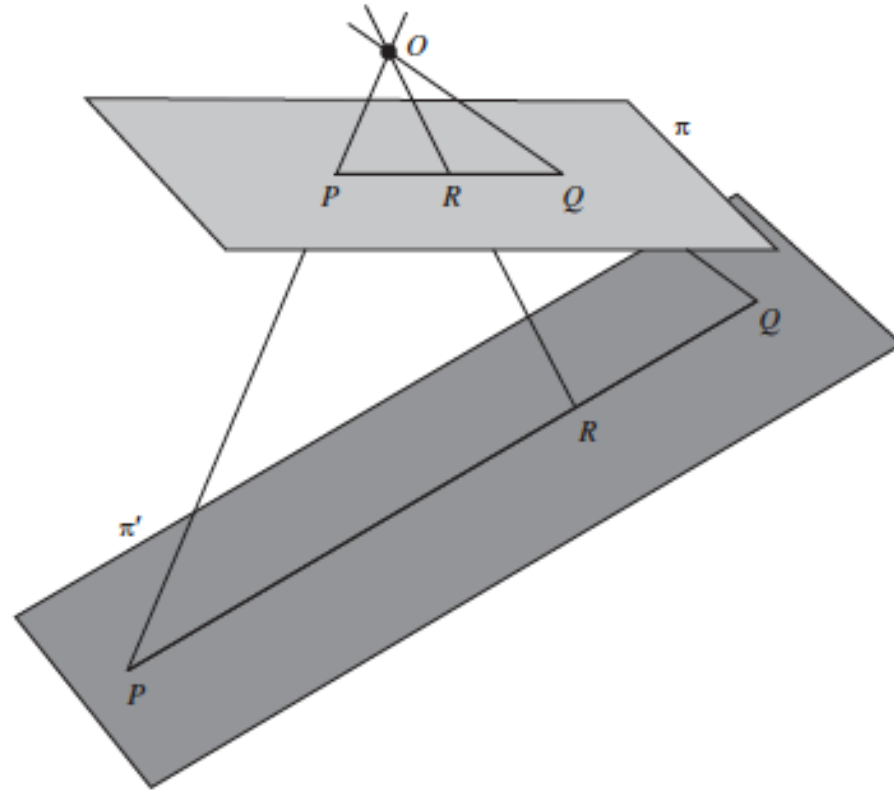
$\approx 1:2$



$\approx 1:2.5$

Clearly, length ratios are not preserved under homographies!

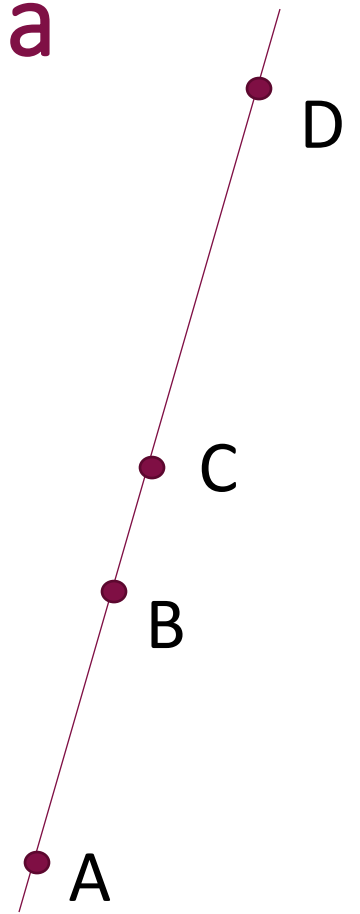
Length ratios under homography



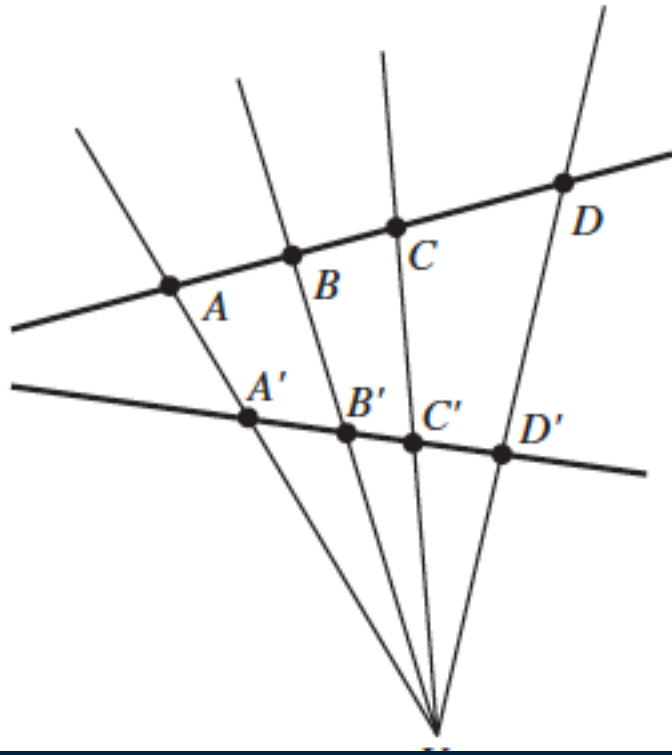
Clearly, length ratios are not preserved under homographies!

What metric property along a line *is preserved* under a projective transformation?

- Not lengths AB i.e. distance of two pts from each other.
- Not length ratios i.e. distance of two pts from a third collinear pt. AC: BC
- Instead, what is preserved is:
 - **Ratio of ratios** of distances of two pts from two other collinear pts.
 - $\frac{AC}{AD} : \frac{BC}{BD} = \text{Cross ratio of A, B, C, D}$
- CR can also be written as $\frac{AC \cdot BD}{AD \cdot BC}$



Cross Ratios of Collinear Points



Given four points A, B, C, D , we define the cross-ratio of their distances as

$$CR(A, B, C, D) = \frac{AC}{AD} : \frac{BC}{BD}.$$

$CR(A, B, C, D)$ remains invariant under projective transformations

$$\frac{AC}{AD} : \frac{BC}{BD} = \frac{A'C'}{A'D'} : \frac{B'C'}{B'D'}$$

Denote $\mathbf{x}'_i = (x'_i, 1)^T$ and $\mathbf{x}_i = (x_i, 1)^T$. Suppose $\mathbf{x}'_i = H\mathbf{x}_i$. Note $|H| = \det(H)$. Then

$$x'_i - x'_j = |\mathbf{x}'_i \ \mathbf{x}'_j| = |\lambda_i H\mathbf{x}_i \ \lambda_j H\mathbf{x}_j| = \lambda_i \lambda_j |H(\mathbf{x}_i \ \mathbf{x}_j)|$$

$$= \lambda_i \lambda_j |H| |\mathbf{x}_i \ \mathbf{x}_j| = \lambda_i \lambda_j \det(H) (x_i - x_j)$$



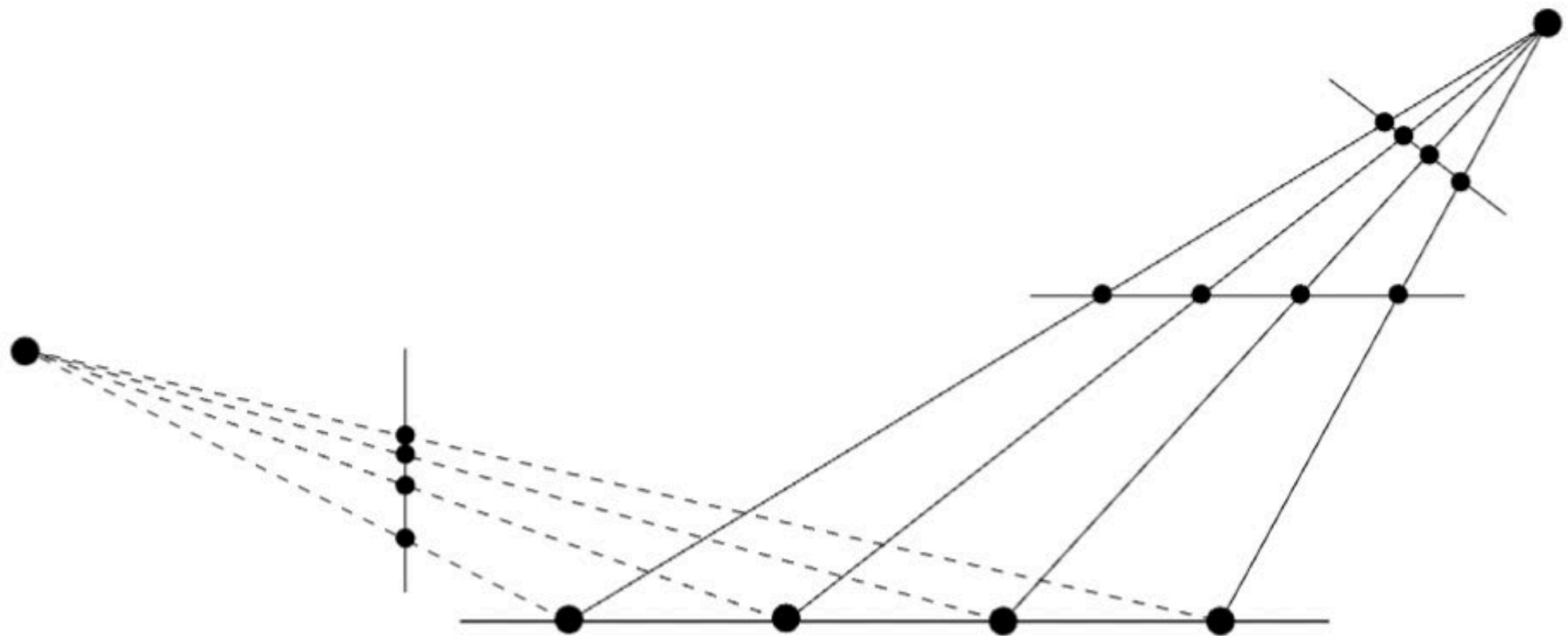
Pappus (290-350 AD)



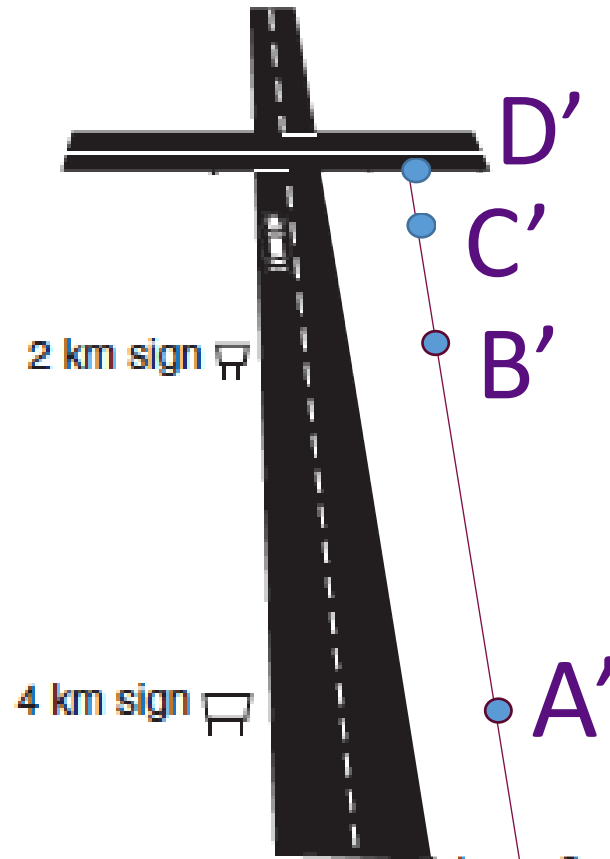
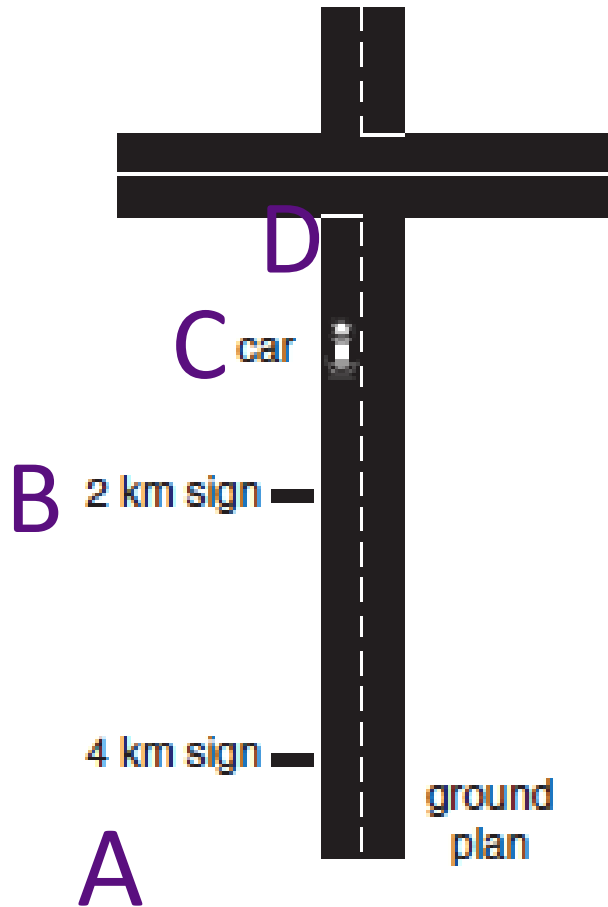
Girard Desargues (1591-1661)

Note: C
but it is
e.g., CR

Same cross ratio for all these 4 point sets!



Example: Cross ratios for metrology



How far is car from intersection:

CD=?

A'D'=300 pixels

A'C'=275 pixels

B'C'=50 pixels

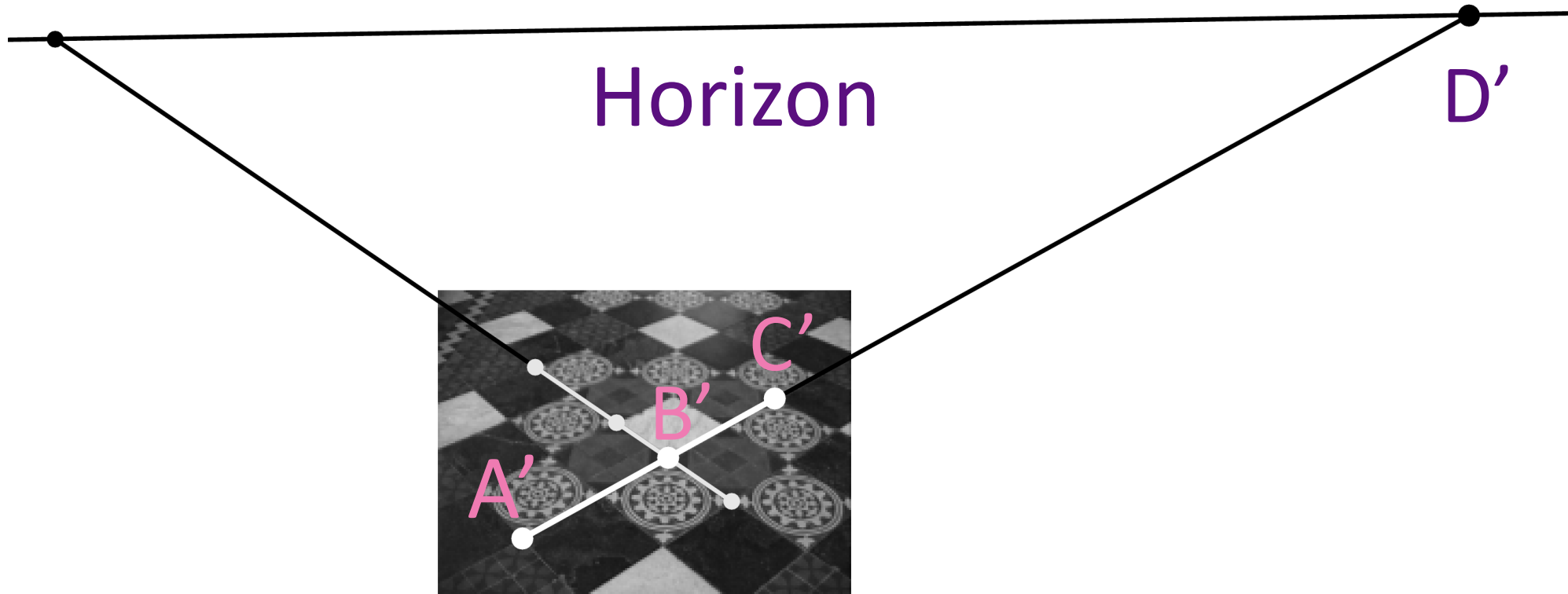
$$\frac{A'C'}{A'D'} : \frac{B'C'}{B'D'} = \frac{275}{300} : \frac{50}{75} = 1.375$$

$$\frac{AC}{AD} : \frac{BC}{BD} = \frac{4-CD}{4} : \frac{2-CD}{2} = 1.375$$

$$CD = 0.857km$$

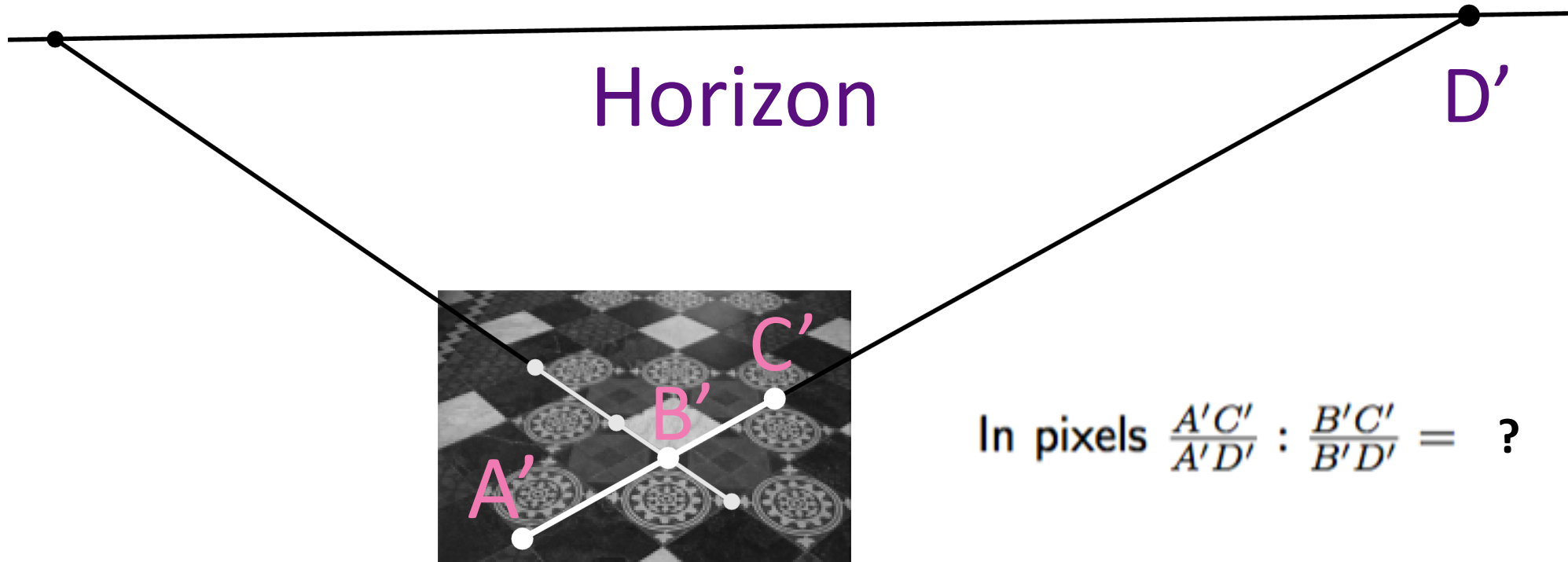
Note: all distances in image are measurable. And 2 distances in world are given.

What happens when one of the points is at infinity?



While D' is a finite point, D on the original plane is at infinity !

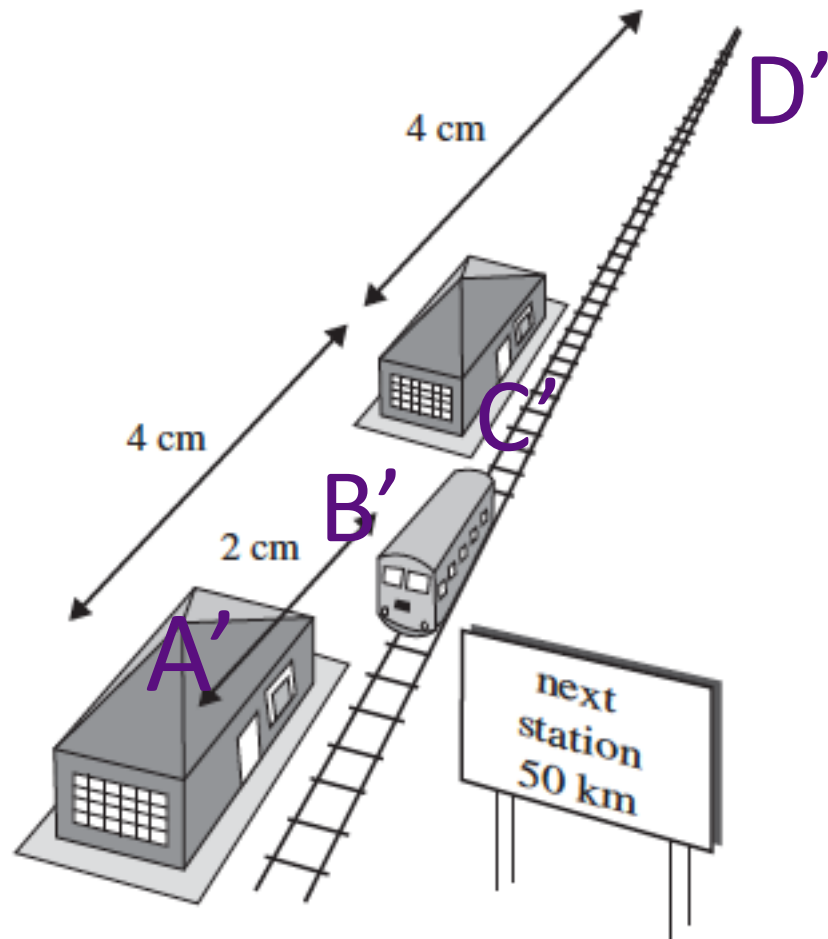
What happens when one of the points is at infinity?



When a point D is at infinity, the cross-ratio becomes a ratio !

$$\frac{AC}{AD} : \frac{BC}{BD} = \frac{AC}{BC} \quad \left(\text{Think } \frac{AC}{\infty} : \frac{BC}{\infty} = \frac{AC}{BC} \times \frac{\infty}{\infty} = \frac{AC}{BC} \right)$$

Vanishing points allow us to measure length ratios!



How far away is the train from the next station?
i.e. Length of BC ?

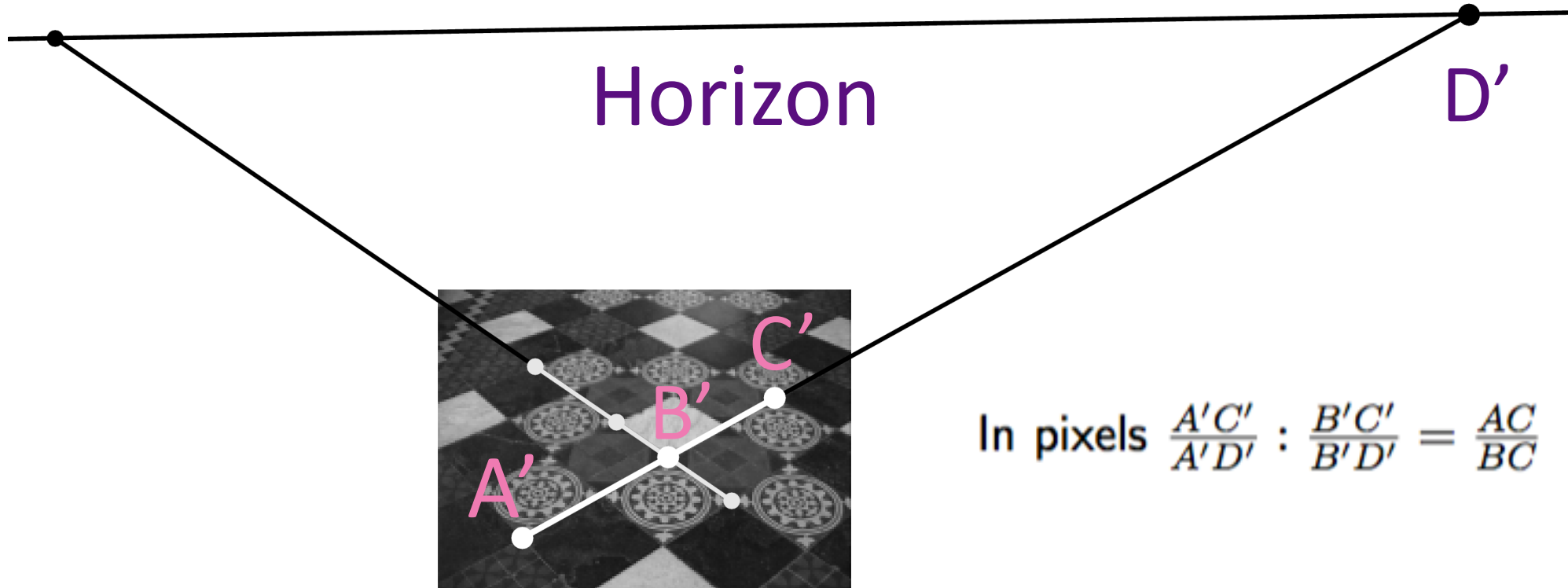
$$\frac{A'C'}{A'D'} : \frac{B'C'}{B'D'} = \frac{AC}{BC}$$

$$\frac{4}{8} : \frac{2}{6} = \frac{50}{BC}$$

$$\frac{4}{8} : \frac{2}{6} = \frac{50}{BC} \text{ and } BC = 33.33 \text{ km.}$$

Note: We only needed measurement of 1 distance in the real world.

Also, world plane length ratios determine vanishing points!



$$\frac{A'C'}{A'D'} : \frac{B'C'}{B'D'} = \frac{AC}{BC} = 2$$

Given: $AB=BC$

If we know A' , B' , C' in pixels, we can find D'

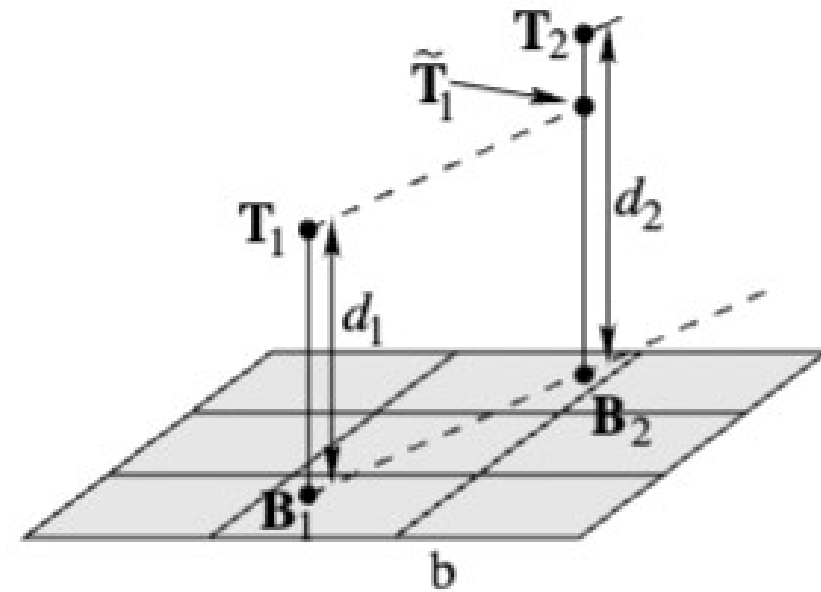
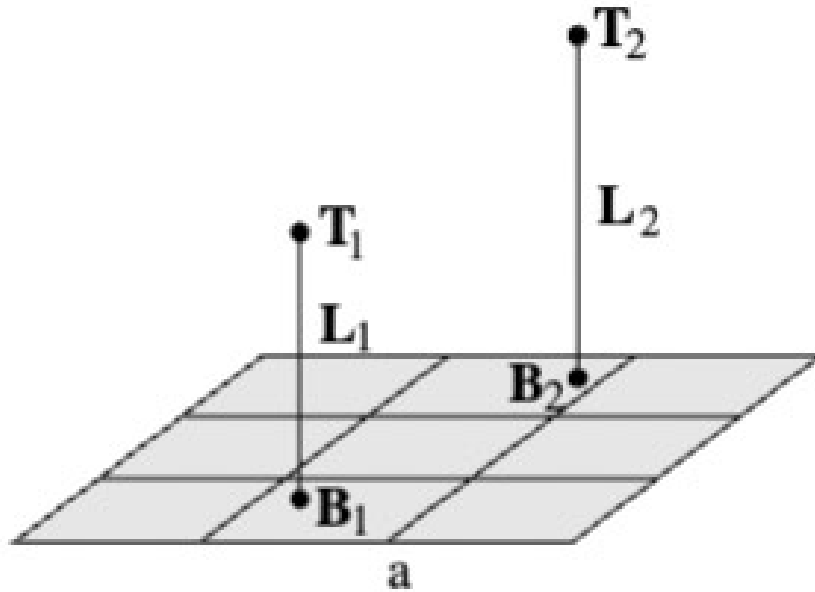
In this way, we can find vanishing points and the horizon without even needing parallel lines!

Measuring heights, i.e. distances *from* a world plane

- So far, we've been looking at distances *on* a world plane.
- Next, distances *off it*?

Length transfer in 3D

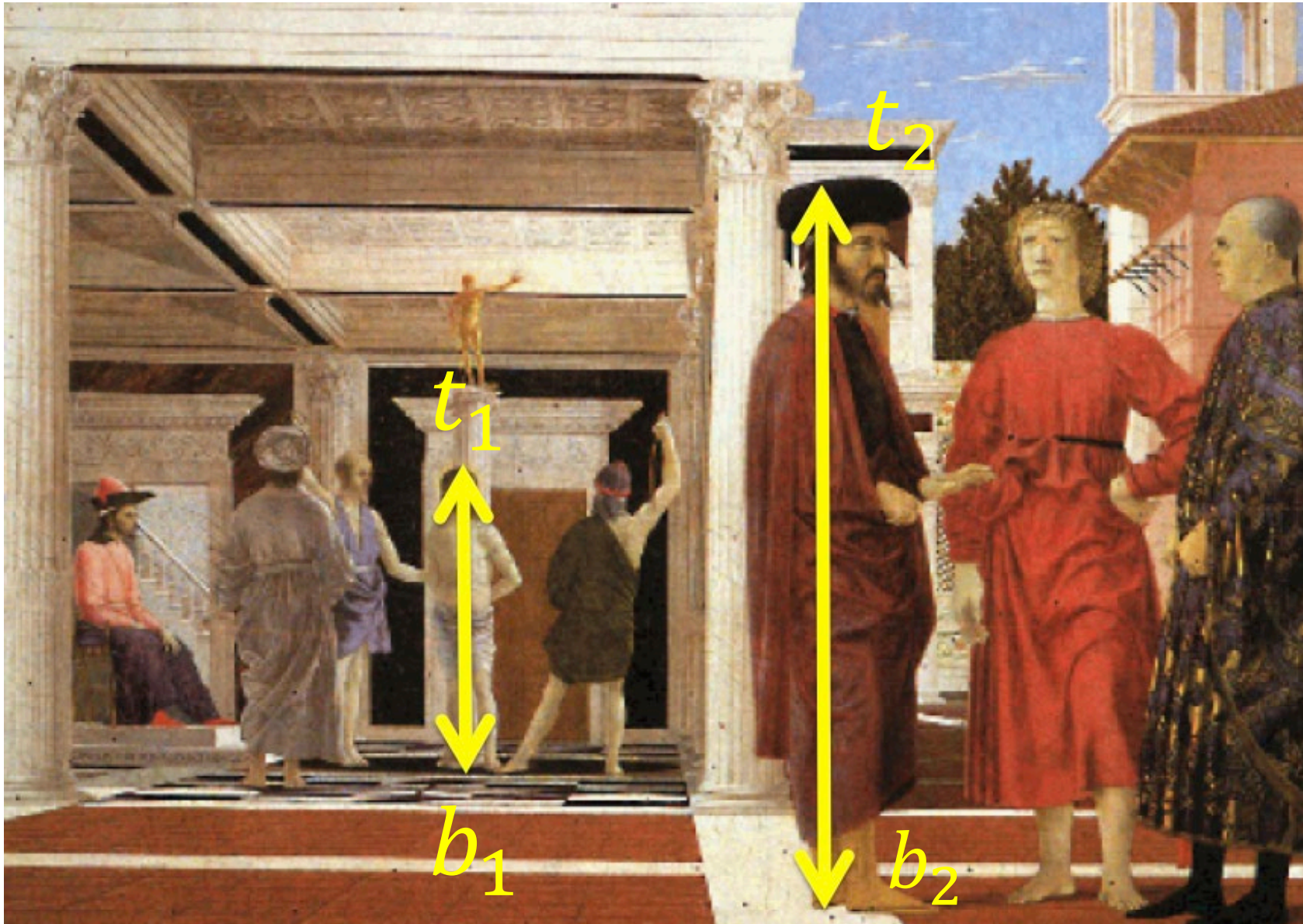
- In the real 3-D world, you can compare one object with known length to another to “transfer” its length. This is what you do with a ruler, for example.



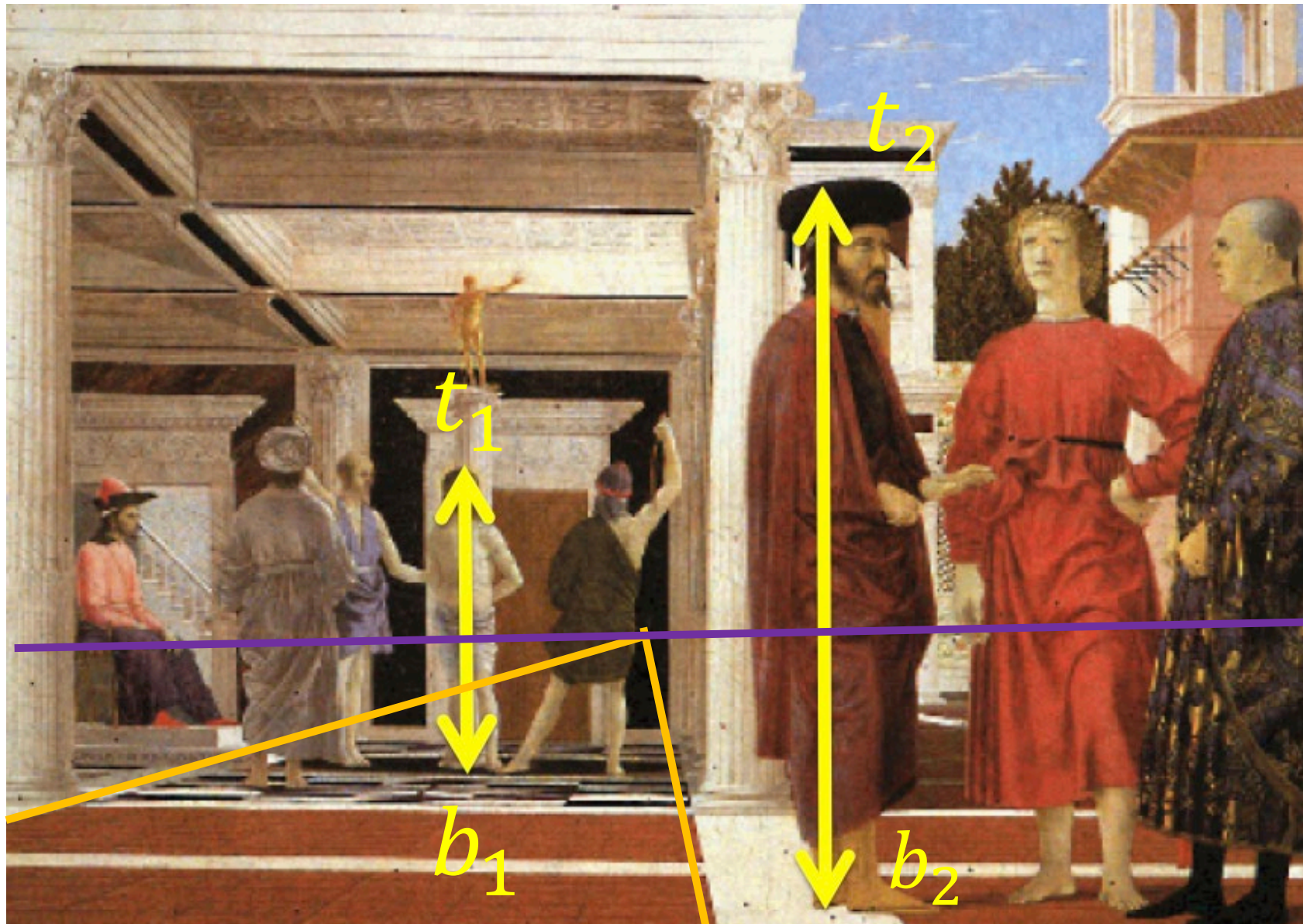
How to do this in an image?

Distance Transfer: How tall is the man if the statue is 180cm (in 3D world space)?

Note: we will be going back and forth between world points (denoted as capital letters) and their projections (denoted as small letters).

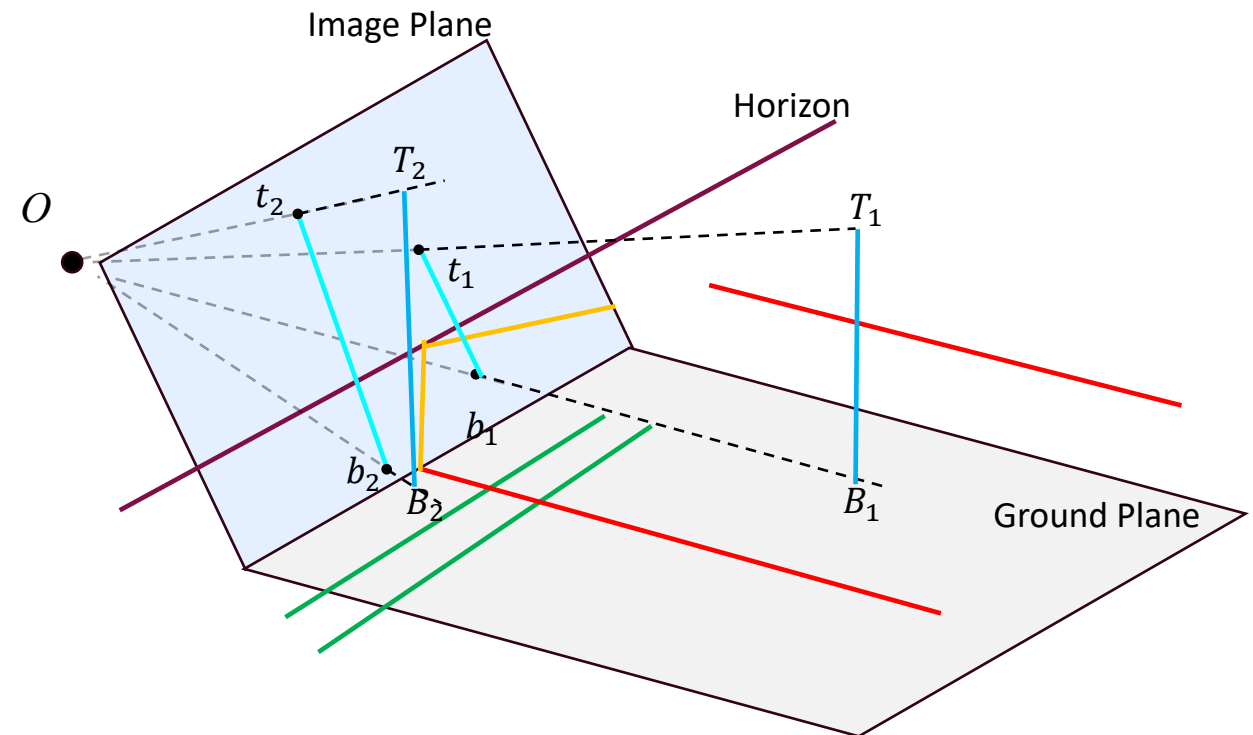
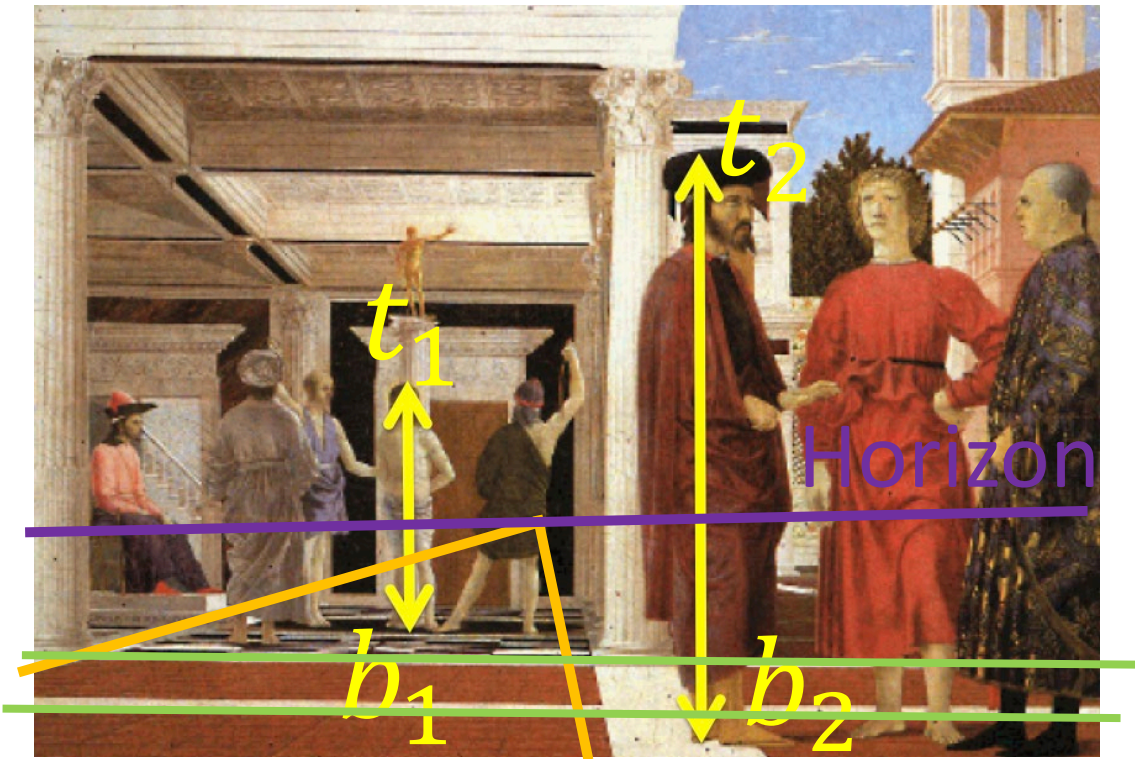


1. Find horizon (from homography, or by finding VPs)

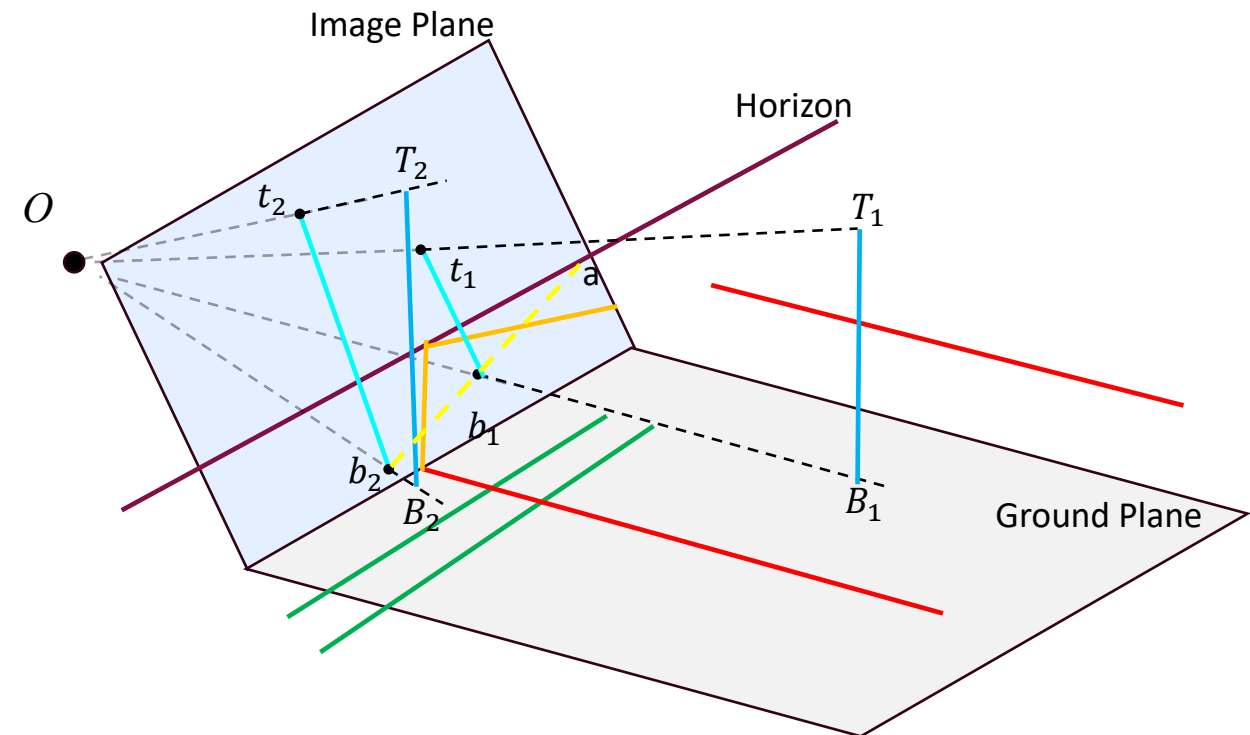
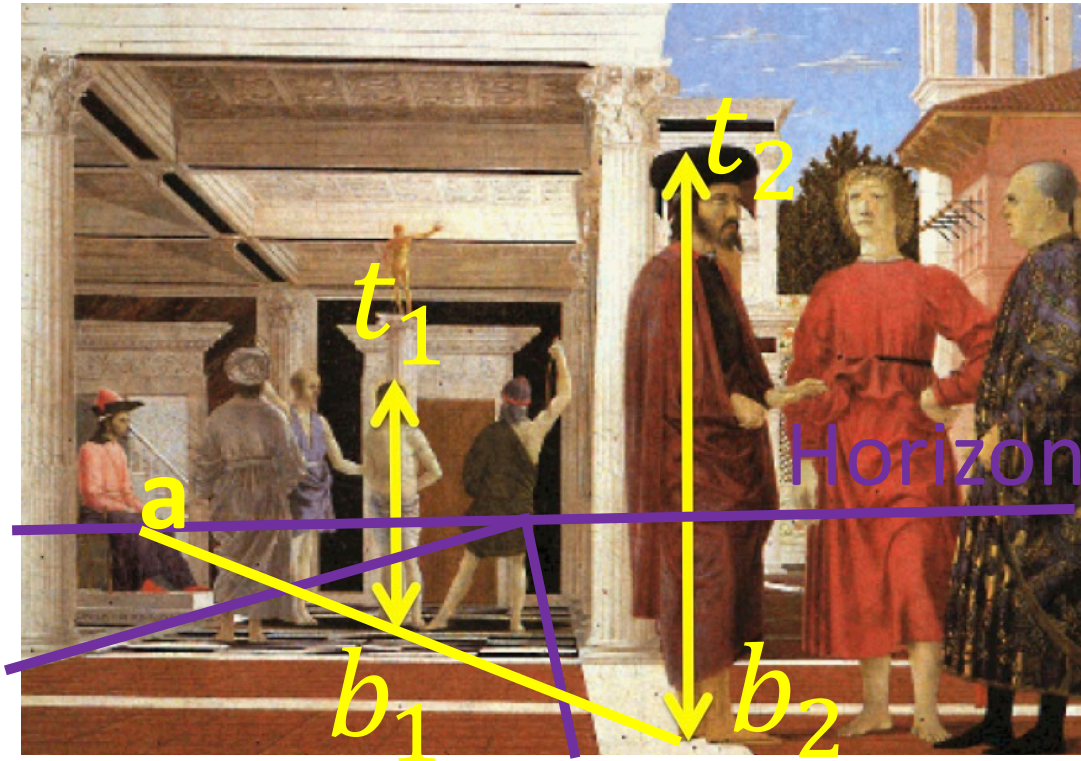


Horizon

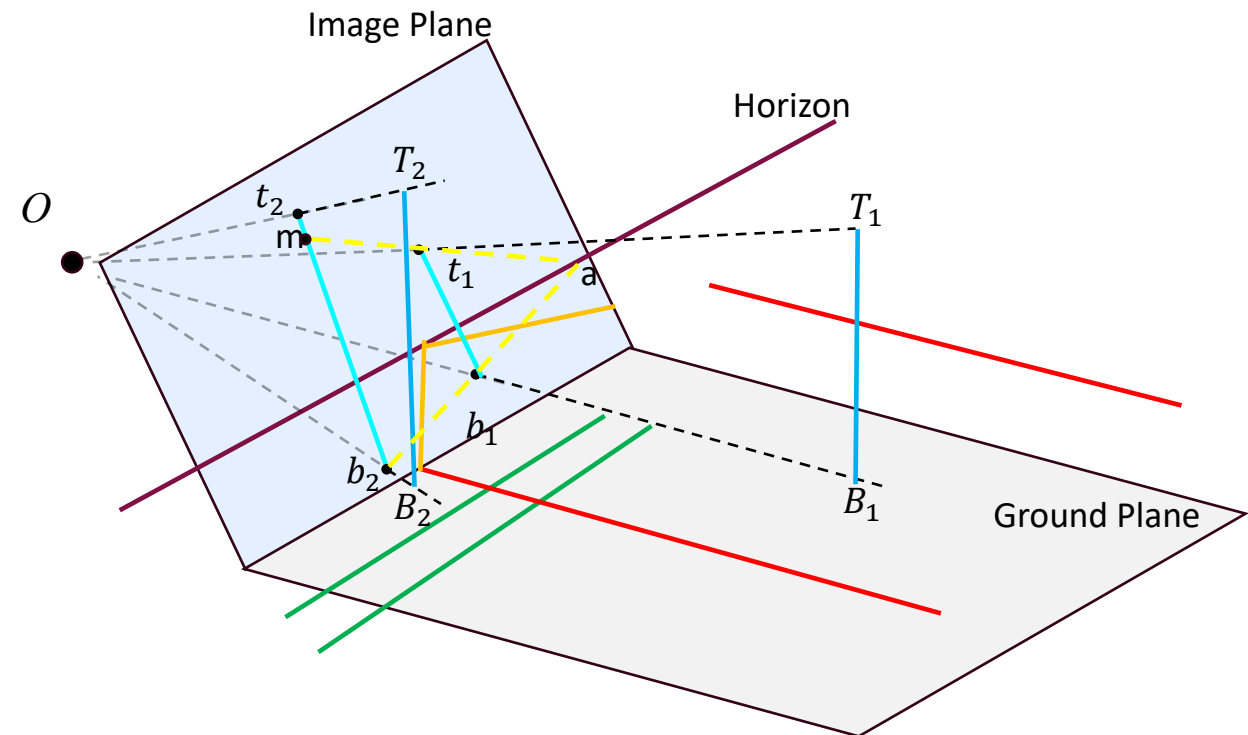
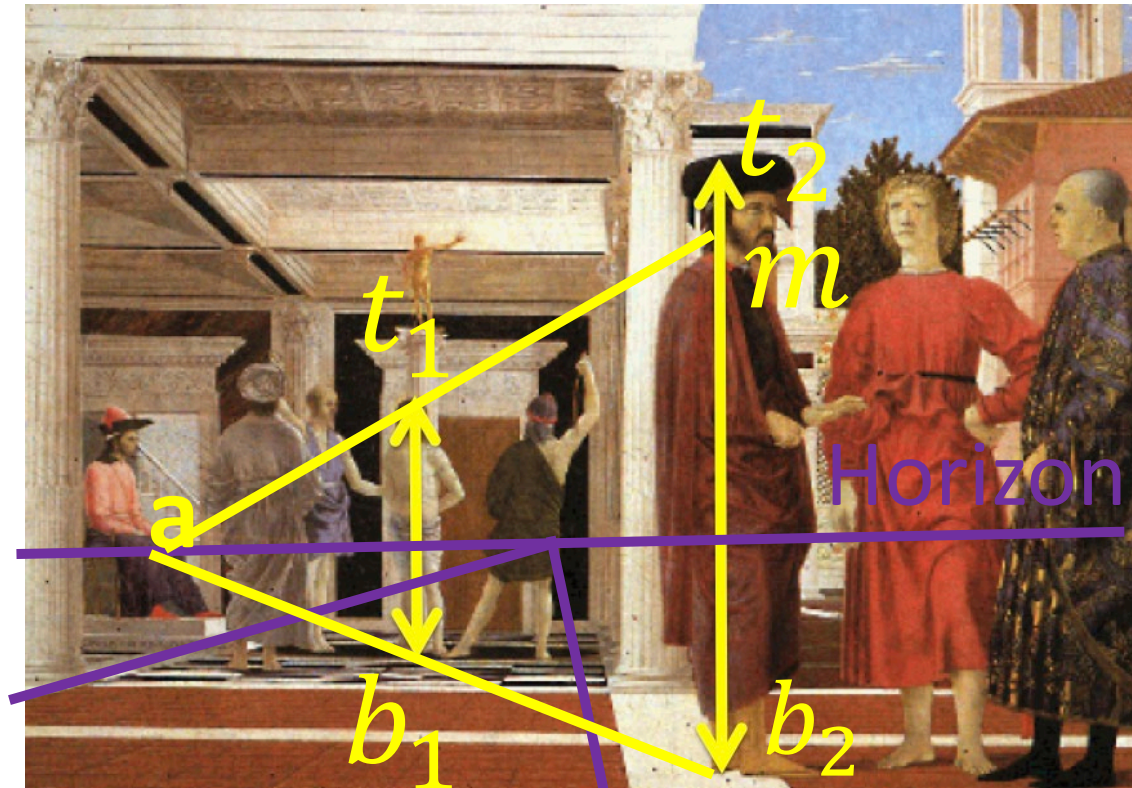
1. Find horizon (from homography, or by finding VPs)



2. Connect the feet of the man and the statue and find intersection a' with horizon!

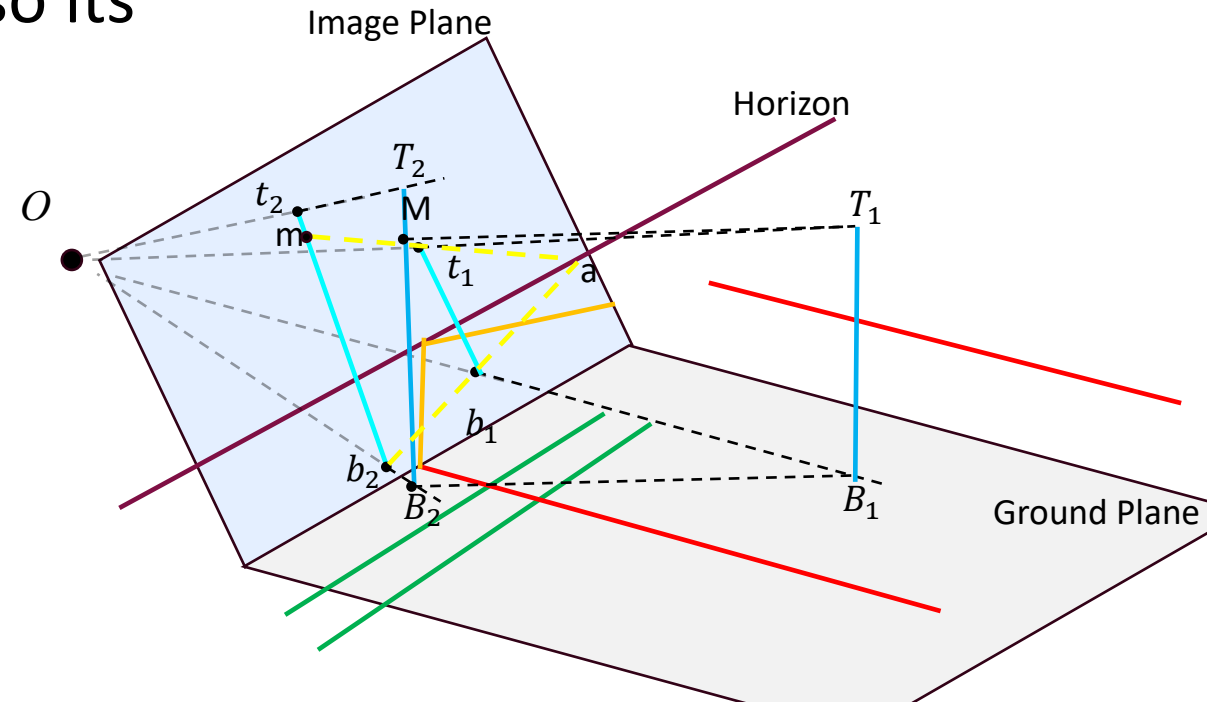


3. Connect a with top of statue t_1

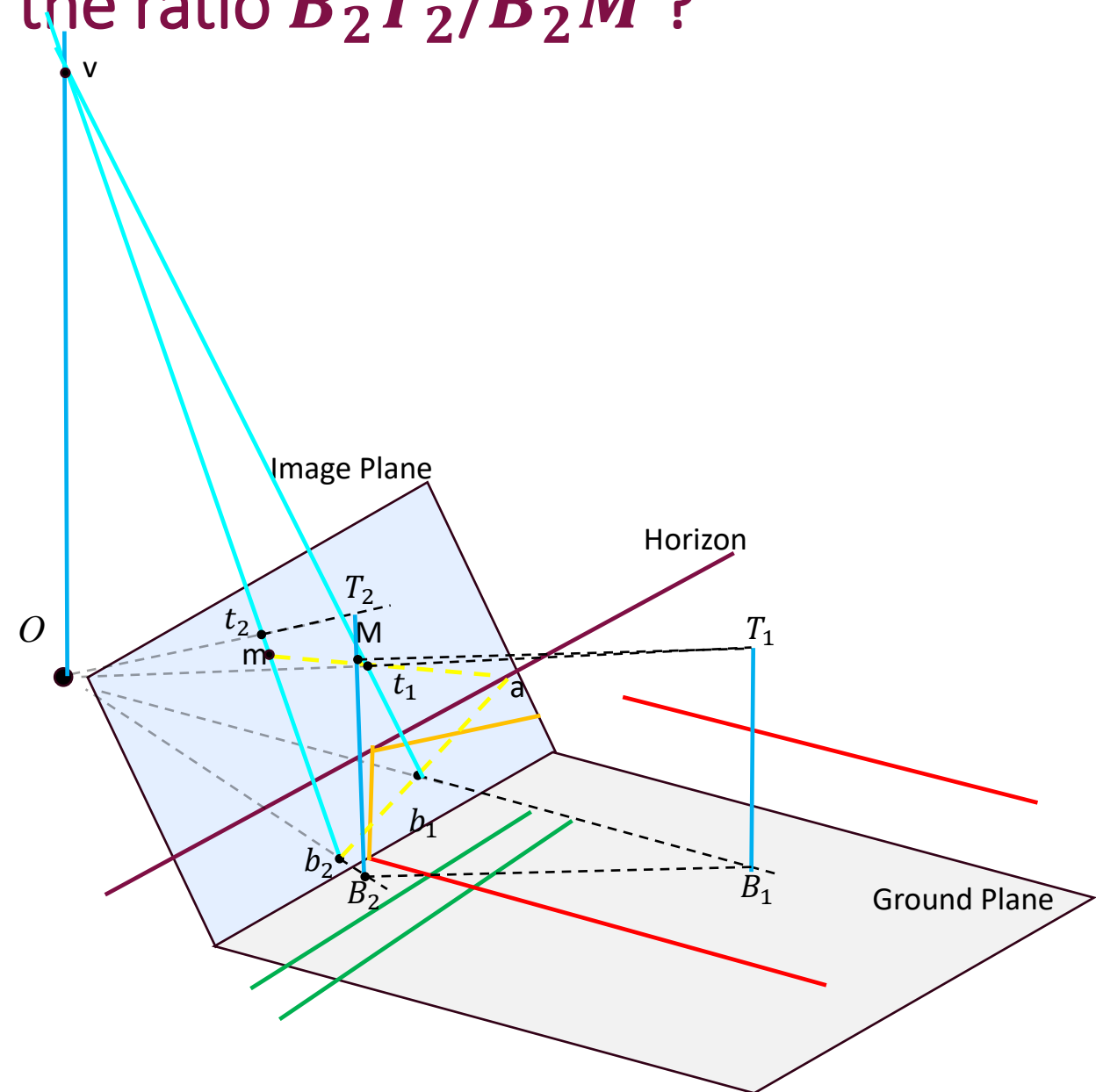
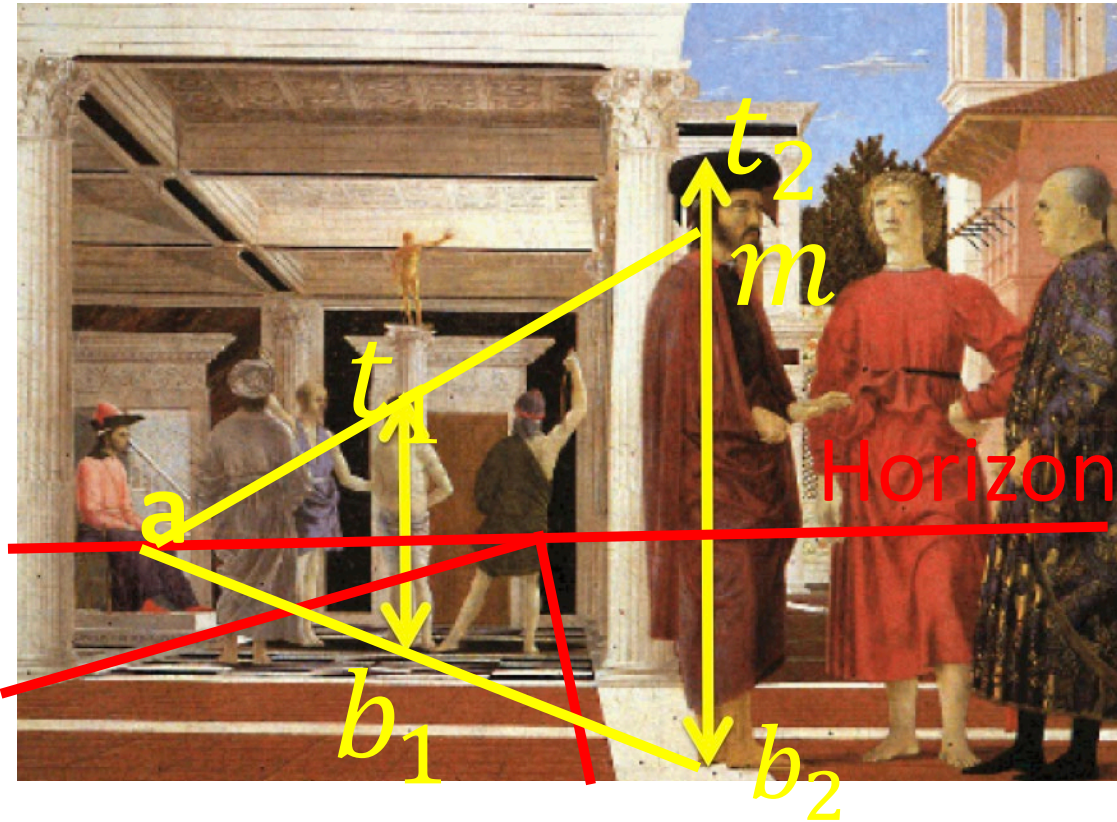


MT_1 is parallel to the ground $\rightarrow B_2M = B_1T_1$

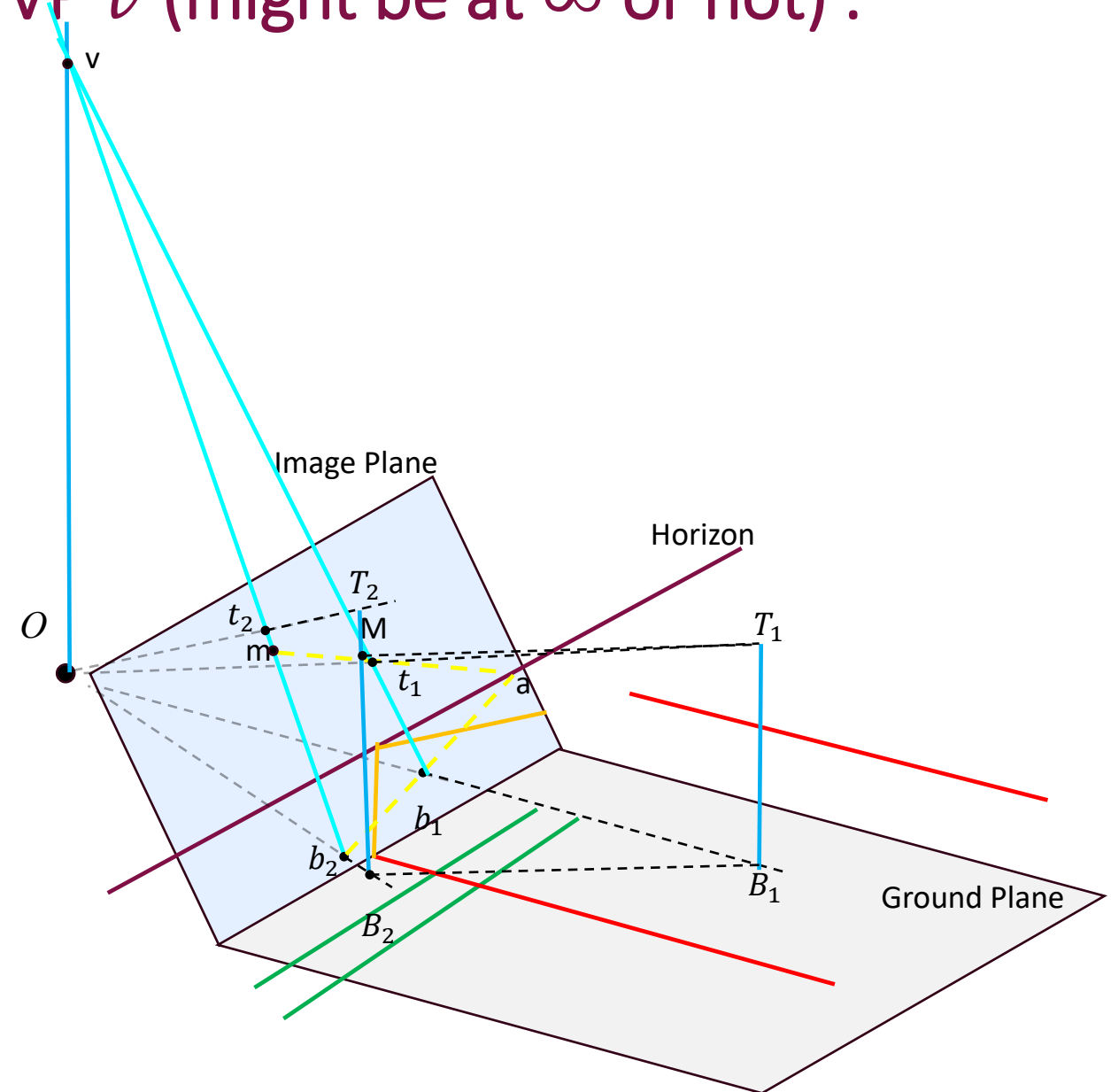
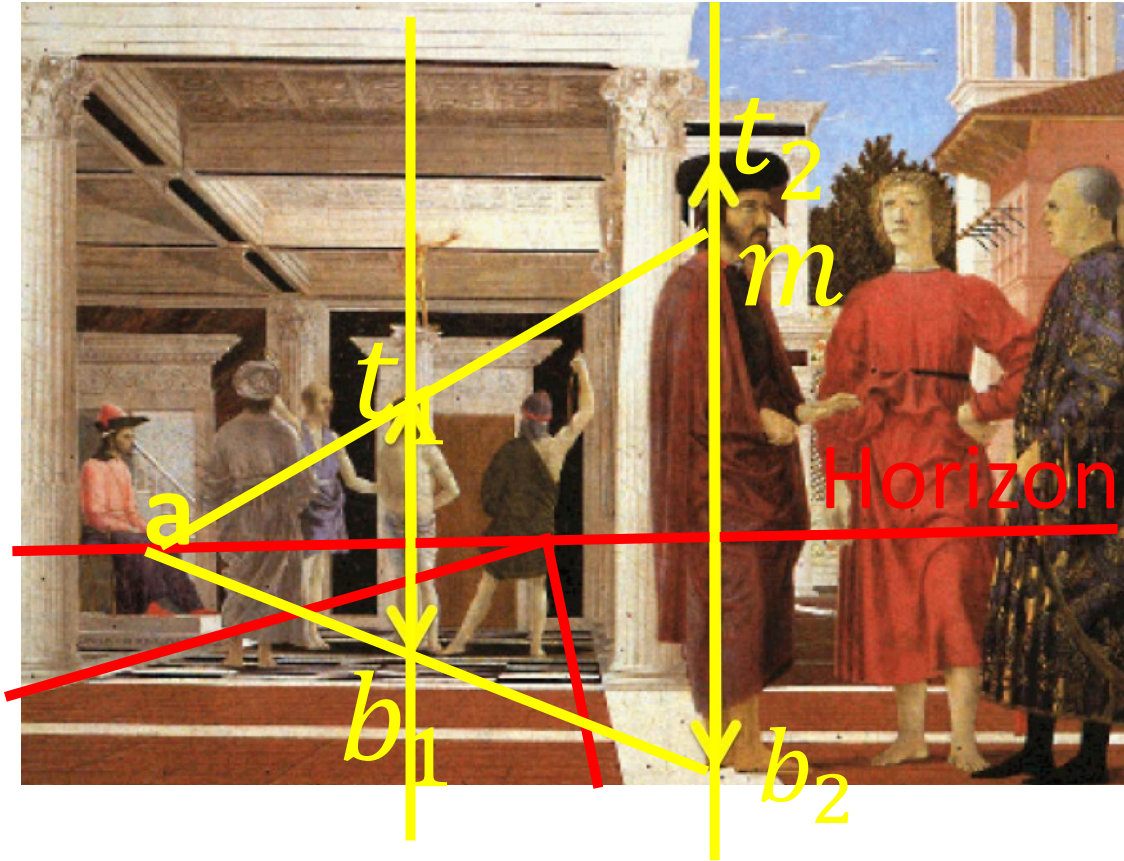
- Consider a world line parallel to B_2B_1 and passing through top of statue T_1 . It intersects the person at a point M that is at the same height as T_1 . (for practical purposes, we will treat the person and the statue as “vertical lines”.)
- The world line MT_1 is parallel to B_2B_1 , so its image mt_1 must meet b_2b_1 at its VP= a .
- So $B_2M = B_1T_1$



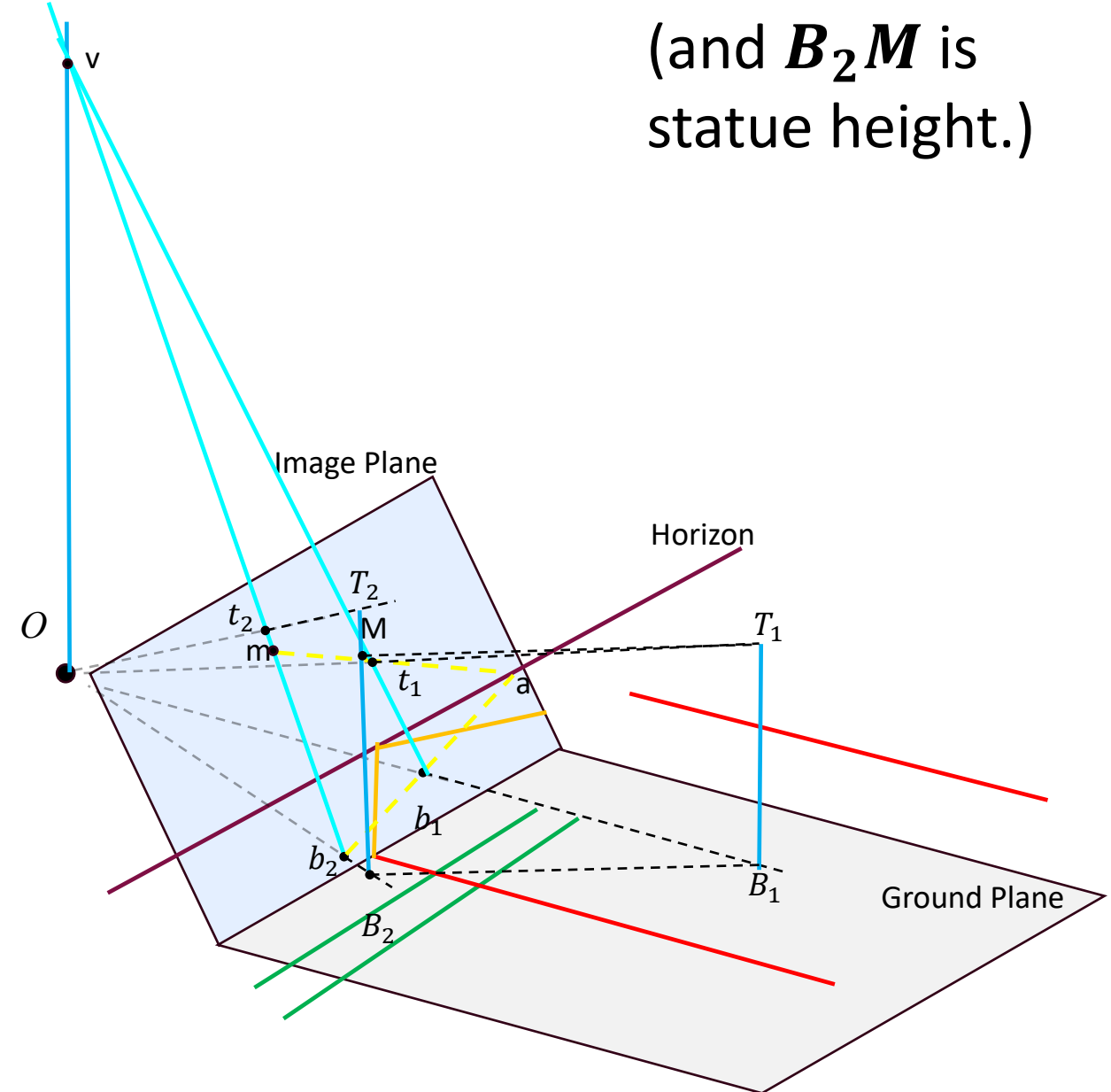
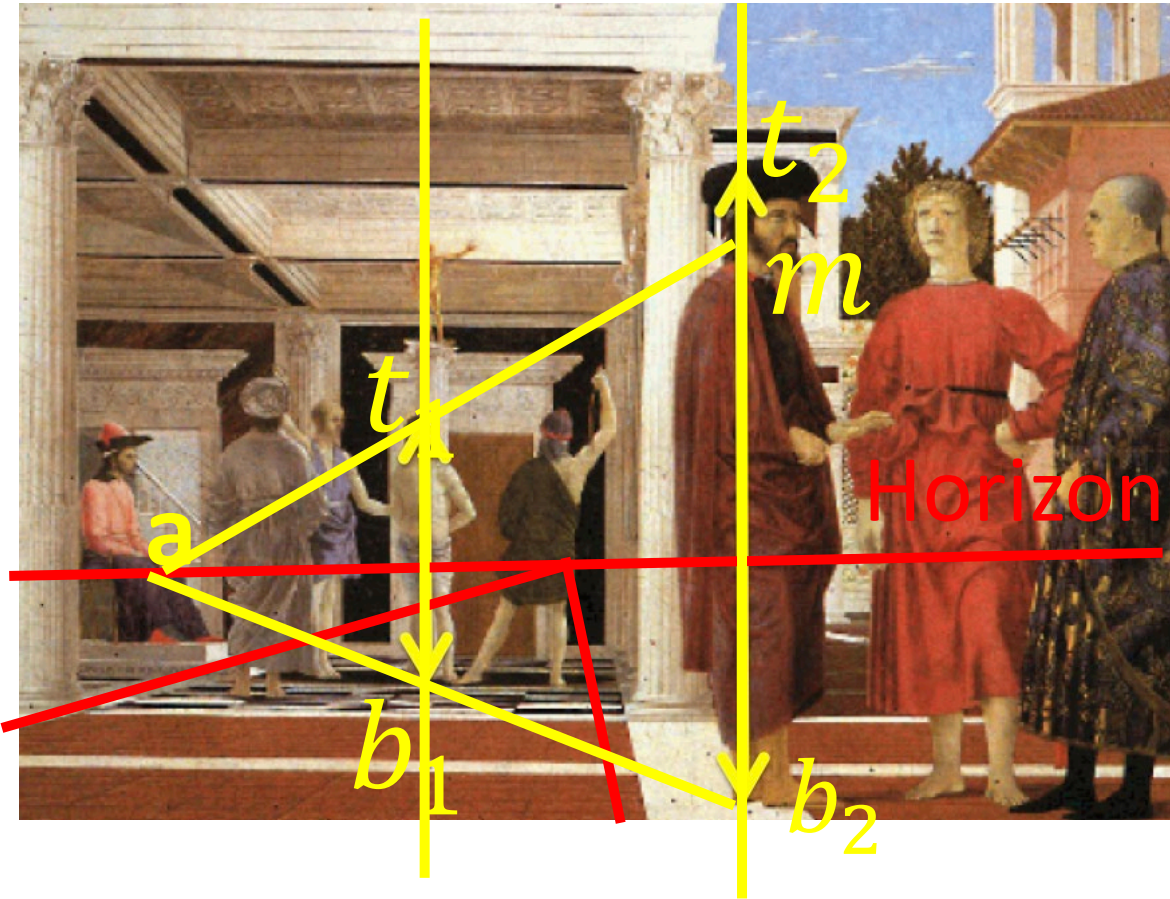
4. But we want B_2T_2 in the world.
How can we compute the ratio B_2T_2/B_2M ?



4. Only if we know a vanishing point in the vertical direction. Let b_1t_1 and b_2m intersect at a vertical VP v (might be at ∞ or not) .



5. Then cross ratio of $\{v, t_2, m, b_2\}$ = ratio $B_2T_2 : B_2M$
 (and B_2M is statue height.)



Single View Metrology via Cross Ratios

- If we know the vanishing point for a direction, we can compute any ratio along this direction!
- We can transfer lengths among parallel line segments in the world using knowledge of the vanishing point for their direction.
- All of this without explicitly computing any focal length, intrinsics, homographies etc.!

We can do “image forensics” on paintings or old photos!

See also: ZH Sec 8.7,
ZH example 8.25

How to Detect Faked Photos

Techniques that analyze the consistency of elements within an image can help to determine whether it is real or manipulated.

